Notes on Abstract Algebra

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1 Logic

Definition If a statement A leads logically to another statement B we say that ‘if A, then B’ and write $A \to B$ which can be thought of as A is *sufficient* for B. If logical space were analogous to sets, we could express ‘if A, then B’ as $A \subseteq B$. Another way to think about implication is that $A \to B$ can be thought of as ‘if A is true, then B must be true’. Yet another way is (‘B, if A’) or that B is *necessary* for A. This can also be expressed as ‘A *only if* B’.

$A \to B$ implies that A being true is sufficient for B to be true. If the we have both $A \to B$ and $B \to A$, then we can say that A is *necessary and sufficient* for B – which can also be expressed as $A \iff B$ or by analogy with set notation as $A \subseteq B \ \ \& \ \ B \subseteq A$.

These logical dependences can be summarized in the statement “*If* is sufficient, *Only If* is necessary.”

> **Necessary Condition:** If A is necessary for B, then B is a subset of A – every time B happens, A must happen.
>
> **Sufficient Condition**: If A is sufficient for B, then A is a subset of B – whenever A happens, B follows.
>
> **Necessary and Sufficient**: A and B are exactly the same set

2 Axioms

Definition The laws of addition for Numbers:

A1: The Commutative Law of addition: $x + y = y + x$ for all pairs of numbers x and y.

A2: The Associative Law of addition: $(x + y) + z = x + (y + z)$ for any three numbers x, y and z.

A3: The existence of a zero. There is a number, called the zero and denoted by 0, which has the property that

$$x + 0 = x \text{ and } 0 + x = x \text{ for all numbers } x.$$

A4: The existence of negatives: Corresponding to each number x there is a number called the negative of x and written $-x$ which satisfies

$$x + -x = 0 \text{ and } -x + x = 0.$$

Definition Distributive Law

D1: The Distributive Law:

$$\begin{aligned} (x+y)z &= xz + yz \\ x(y+z) &= xy + xz \end{aligned}$$

Definition The laws of multiplication:

M1: The Commutative Law of multiplication : $xy = yx$ for all pairs of numbers x and y.

M2: The Associative Law of multiplication: $(xy)z = x(yz)$ for any three numbers x, y and z.

M3: The existence of a unity. There is a number, called the unity and denoted by 1, which has the property that

$$x1 = x \text{ and } 1x = x \text{ for all numbers } x.$$

M4: The existence of inverses: Corresponding to each number x there is a number called the inverse of x and written x^{-1} which satisfies

$$xx^{-1} = 1 \text{ and } x^{-1}x = 1.$$

Definition The Distinction of Numbers

X1: $1 \neq 0$.

$a \cdot 1 = a$ for all $a \in D$ where D represents a suitable domain of numbers.

$a \cdot 0 = 0$ for all $a \in D$.

3 Equivalence Relations

Theorem 3.1 *The equivalence classes theorem*
Suppose in a set S we have a relation R defined between certain pairs of elements, xRy meaning that x and y stand in the given relation R.
Suppose further that R has the following 3 properties:

(1) It is reflexive: i.e. xRx for all $x \in S$.

(2) It is symmetric: i.e. $xRy \Rightarrow yRx$.

(3) It is transitive: i.e. xRy and $yRz \Rightarrow xRz$.

Then R is an equivalence relations: i.e. it divides S into mutually exclusive subsets so that every element of S is in one and only one subset, and so that two elements are in the same subset if and only if they stand in the relation R to one another.

The subsets are called equivalence classes defined by R.

Given any element x in S consider all the elements y such that xRy. These form a subset of S: let us call it A_x. We show first that two elements are in the same subset A_x if and only if they stand in the relation R to one another. Suppose yRz and $y \in A_x$. Then xRy and so xRz since R is transitive. Hence $z \in A_x$. Conversely, if y and z are both in A_x we have xRy and xRz, i.e. yRx by the symmetric property and xRz: thus yRz by transitivity.

For each element x we now have a subset A_x, but these will not all be distinct. We will show that two such are either mutually exclusive or else identical. Suppose A_x and A_y both have an element z, Then xRz and yRz, and so zRy by the symmetric property; hence xRy by the transitive property. Now take any element w of A_x. Then xRw and since xRy we have yRx, giving yRw. So w is in A_y. Hence we have shown that if A_x and A_y have one element z in common, any element w of A_x is in A_y, i.e, $A_x \subset A_y$. Similarly $A_y \subset A_x$, and so the subsets A_x and A_y are identical.

Thus we have mutually exclusive subsets $A_{x_1}, A_{x_2}, ...$ Finally by the reflexive property any element t is in one of the subsets, viz. A_t.

4 Groups, Subgroups and Cosets

Abstract Group

A set of elements S forms a group if to any two elements x and y of S taken in a particular order there is associated a unique third element of S, called their product and denoted by xy, which satisfies the following laws

(M2) For any three elements

$$x, y \text{ and } z: \quad (xy)z = x(yz).$$

(M3) There is an element called the *neutral element* and denoted by e which has the property that $xe = ex = x$ for all elements x.

(M4) Corresponding to each element x there is an element x^{-1} called the *inverse* of x which has the property that

$$xx^{-1} = x^{-1}x = e.$$

Subgroups

A non-empty subset H of a group G is a subgroup of G if:

(a) Given any ordered pair (h_1, h_2) of elements of H the product h_1h_2 as defined in G is in H.

(b) The neutral element of G is in H.

(c) Given any element h in H, the inverse h^{-1} as defined in G is in H.

This a subset of elements of G is a subgroup, if and only if it forms a group, under the same group product.

Note that (b) is not needed, since if $h \in H, h^{-1} \in H$ by (c) and so $hh^{-1} = e \in H$ by (a).

Theorem 4.1 *A necessary and sufficient condition for a non-empty subset H to be a subgroup of a group G is that $gh^{-1} \in H$ for all $g, h \in H$.*

(1) Necessary: If H is a subgroup $h^{-1} \in H$ and so $gh^{-1} \in H$.

(2) Sufficient: If g is any element of H we have $gg^{-1} \in H$, i.e, $e \in H$. Hence $eg^{-1} = g^{-1} \in H$, i.e, H contains inverses. Thus if $g, h \in H$, so is h^{-1} and hence $g(h^{-1})^{-1} \in H$, i.e. $gh \in H$.

Given a group G and a subgroup H it is possible to decompose the whole group into subsets 'parallel' to H, two elements being in the same subset if their 'difference' is in H. This concept is better adapted in the product notation where the difference between g and k is expressed as $k^{-1}g$. We therefore decompose group G into subsets such that g and k are in the same subset if $k^{-1}g \in H$. These subsets are known as cosets and we have a true decomposition into mutually exclusive subsets because $k^{-1}g \in H$ sets up an equivalence relation gRk if $k^{-1}g \in H$.

Theorem 4.2 *Equivalence Classes of G relative to H The relation defined by gRk above is an equivalence relation.*

(1) Reflexive: For any $g \in H$ we have gRg because $g^{-1}g = e \in H$ since H is a subgroup.

(2) Symmetric: If gRk then $k^{-1}g \in H$. Hence its inverse is in H, i.e. $g^{-1}k \in H$ and so kRq.

(3) Transitive: If gRk and kRl then $k^{-1}g \in H$ and $l^{-1}k \in H$. Hence $(l^{-1}k)(k^{-1}g) = l^{-1}g \in H$ and so gRl.

These mutual exclusive classes are called left cosets of G relative to H. So if gRk then kRg and $g^{-1}k \in H$ or $k = gh$ for some $h \in H$. Hence the coset containing g is precisely the complex gH, the set $\{gh\}$ for all $h \in H$.

Theorem 4.3 *Left Cosets The left cosets of G relative to H are the complexes gH, $g \in G$. Any left coset may be expressed in this form for any g in it. Any two cosets are either the same of have no element in common.*

We may similarly define *right cosets* of G relative to H as the equivalence classes under the relation given by $gk^{-1} \in H$. They are the complexes Hg.

We normally will use left cosets and therefore will just refer to them as cosets.

5 Mappings and Homomorphisms

Definition An injective function $f : X \to Y$ (also known as injection, or one-to-one function) is a function f that maps distinct elements of its domain to distinct elements; that is, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Equivalently, $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$ in the equivalent contrapositive statement.

Definition A surjective function $f : X \to Y$ (also known as surjection, or onto function) is a function f that every element y can be mapped from some element x so that $f(x) = y$. In other words, every element of the function's codomain is the image of at least one element of its domain. It is not required that x be unique; the function f may map one or more elements of the set X to the same element of the set Y.

Definition Homorphisms and Mappings:
Monomorphism: 1-1, Epimorphism: onto, Isomorphism: 1-1 and onto.
Homomorphisms preserve structure (i.e. multiplication and addition in Groups and Rings).
Homomorphisms that have the same object and image space are called Endomorphisms.
And Endomorphism that is also an Isomorphism is called and Automorphism.
In a Group G, the mappings of G into itself, given by $g \to x^{-1}gx$, where x is any fixed element of G, is an Automorphism, known as an Inner Automorphism.

Suppose $f : A \to B$ and $g : B \to C$. Hence $(g \circ f) : A \to C$ exists. Show

(a) If f and g are 1-1 (monomorphisms), then $g \circ f$ is 1-1:
Suppose $(g \circ f)(x) = (g \circ f)(y)$. Then $g(f(x)) = g(f(y))$. Because g is 1-1, $f(x) = f(y)$ and because f is 1-1, $x = y$. Therefore $(g \circ f)(x) = (g \circ f)(y)$, implies $x = y$; hence $g \circ f$ is 1-1.

(b) If f and g are onto (epimorphisms), then $g \circ f$ is onto.
Suppose $c \in C$. Because g is onto, there exists $b \in B$ for which $g(b) = c$. Because f is onto, there exists $a \in A$ for which $f(a) = b$. Thus $(g \circ f)(a) = g(f(a)) = g(b) = c$, Hence $g \circ f$ is onto.

(c) If $g \circ f$ is 1-1, then f is 1-1.
Prove contrapositive, Suppose f is not 1-1., Then there exists distinct elements $x, y \in A$ for which $f(x) = f(y)$. Then $(g \circ f)(x) = g(f(x)) = g(f(y)) = (g \circ f)(y)$. Hence $g \circ f$ is not 1-1.Therefore if $g \circ f$ is 1-1, then f must be 1-1.

(d) If $g \circ f$ is onto, then g is onto.
Prove contrapositive, If $a \in A$, then $(g \circ f)(a) = g(f(a)) \in g(B)$. Hence $(g \circ f)(A) \subseteq g(B)$. Suppose g is not onto. Then $g(B)$ is properly contained in C and so $(g \circ f)(A)$ is properly contained in C; thus $g \circ f$ is not onto. Therefore if $g \circ f$ is onto, then g must be onto.

Definition A is called the object space of the mapping, B is the image space, a is the object whose image is b and we say that we have a mapping of A into B. If the mapping is denoted by θ, we write $\theta : A \to B$, and we say that $a\theta = b$ or $a \to b$.

In any homomorphism between a group G and another group H the neutral element e of G maps into the neutral element f of H. The set of elements which form a subset of G which map into f is called the *kernel* of the homomorphism.

Theorem 5.1 *The kernel of any homomorphism is a subgroup of G.*

Theorem 5.2 *The image $G\theta$ is a subgroup of H.*

6 Rings

A set R or elements forms a ring if to any two elements x and y of R taken in a particular order there are associated elements of R, called their sum and product and denoted by $x + y$ and xy respectively, both being unique, and such that axioms $A1, A2, A3, A4$ and $M2$ and $D1$ above are satisfied.

Basic types of rings
Commutiative rings
The most obvious additional axiom is that of commutativity of multiplication $M!$, that $xy = yx$ for all x and y in the ring. Such a ring is called a *commutative ring* (addition is always commutative).

Rings with unity A ring admits of multiplication but not necessarily of division. For the latter to be possible we must have a neutral element, and also inverses, which are defined in terms of the neutral element. If both these are present, the structure becomes a *field*. However, a ring may have a neutral element and yet not possess inverses (example – the ring of integers). Such a rings is called a *ring with unity*.

7 Ideals

Definition A subring I of a ring R is a two-sided ideal) if for all $x \in R$ and $i \in I, xi$ and ix are both in I.

Note that because R may not be a commutative ring we must have both xi and ix in I.

Theorem 7.1 *If R has a unity then*

$$\{xi : x \in R, i \in I\} \subseteq I \implies \{xi\} = I \text{ and similarly for } \{ix\}.$$

Proof $\{xi\} \supseteq \{1i\} = I$

Quotient Rings

Theorem 7.2 *If I is an ideal of a ring R the cosets $x + I$ form a ring under the sum $(x+I)+(y+I) = (x+y)+I$ and products $(x + I)(y + I) = xy + I$.*

The ring of cosets is called the *Quotient Ring of I in R* and it is written R/I. The mapping $a \to a' = a + I$ for $a \in R$, which carries each element of R into the coset containing it is a homomorphism of R into the quotient ring R/I and the elements mapped on zero in this homomorphism are exactly the elements of the ideal I. In the homomorphic ring, the zero is the coset $0 + I$, hence the elements of I are mapped onto the zero coset.

Theorem 7.3 *If a homomorphism H maps R on R' and exactly the elements of I on zero, then R' is isomorphic to the quotient ring R/I.*

If b is an element of a commutative ring with unity, the set $[b]$ of all multiples xb of b, for any $x \in R$, is an ideal known as a *principal ideal*; it is the smallest ideal of R containing b.

Inclusion between ideals is closely related to divisibility between numbers. In the ring J it integers $n|m$ means that $m = an$, for some integer a (n divides m). Hence, every multiple of m is a multiple of n. The multiples of n constitute the principle ideal $[n]$, so the result means that $[m]$ is contained in $[n]$. Conversely $[m] \subseteq [n]$ means in particular that m is in $[n]$, and hence $m = an$. Therefore

$$[m] \subseteq [n] \text{ if and only if } n|m.$$

The "bigger" number corresponds to the "smaller" ideal; for instance the ideal [6] of all multiples of 6 is properly contained in the ideal [2] of all even integers.

8 Fields

Theorem 8.1 *(Fields have no non-trivial Ideals): If a field F has an ideal I, then $I = (0)$ or $I = (1) = F$.*

Proof Let $a \in I$ and $a \neq 0$. Then $aa^{-1} = 1 \in I$ and so $f.1 = f \in I$ for any $f \in F$ Therefore $F \subset I$. But $I \subset F$ by definition of Ideals. Therefore $I = F$. The only other possibility is $a = 0$ in which case we have the trivial Ideal $I = (0) = 0$.

If F is any field, then the smallest subfield of F that contains the identify element 1 is called the *prime subfield* of F. If F is a finite field, then its prime subfield is isomorphic to $\mathbb{Z}_p$, where $p = \mathrm{char}(F)$ for some prime p. When we want to emphasize $\mathbb{Z}/p\mathbb{Z}$ with its field structure, we denote it by $\mathbb{F}_p$. The prime subfield of $\mathbb{Z}_p$ which is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ is $\mathbb{F}_p$. The only prime fields are the fields of residues modulo p and the field of rationals.

The prime subfield of a field F is the intersection of all subfields of F. The intersection of any two subfields is a field and similarly for all the other subfields. Also the intersection of all subfields has no proper subfields, since any such would be subfields of R and hence must contain the intersection which is impossible. Hence the intersection is a prime subfield. F cannot contain two distinct prime subfields, since their intersection would also be a subfield and hence must be identical with both.

Theorem 8.2 *a Field has no proper homomorphic images*

Proof It is sufficient to show that a field, considered as a ring, can have no proper ideals.

Theorem 8.3 *Any non-empty subset I of polynomials $F[x]$ which contains the sum and difference of any two, and all the multiples of any one of its members, consists either: 1) of 0 alone; or 2) of the set of multiples $q(x)a(x)$ of any non-zero member $a(x)$ of the least degree.*

Proof Unless $I = 0$, it contains a non-zero polynomial $a(x)$ of least degree $d(a)$ and, with $a(x)$, all its multiples $q(x)a(x)$. In this case, if $b(x)$ is any polynomial in I, using the division algorithm there exists some polynomial $r(x) = b(x) - q(x)a(x)$ which has degree less than $d(a)$. But by hypothesis I contains $r(x)$ and by construction no non-zero polynomial of degree less than $d(a)$. Hence $r(x)$ is the zero polynomial and $b(x) = q(x)a(x)$ ∎

Using the above result of Thm. (8.3) we have in the domain $F[x]$ of polynomials $f(x)$ over F every ideal is a principal ideal.

9 Integral Domain

An integral domain is a commutative ring with a unity that has no zero divisors: $xy = 0 \to x = 0$ or $y = 0$.

Theorem 9.1 *The absence of zero divisors is equivalent to the Cancellation Law.*

If $xy = 0 \to x = 0$ *or* $y = 0$*, then*

$$xy = xz \to y = z \text{ or } x = 0,$$

and conversely.

Proof Suppose first that there are no zero divisors, Then if

$$xy = xz,\ xy - xz = 0 \text{ and so } x(y - z) = 0.$$

Hence either $x = 0$ or $y - z = 0$ by hypothesis, i.e., $x = 0$ or $y = z$.

Conversely let the Cancellation Law be true and suppose that $xy = 0$. Then since $x0 = 0, xy = x0$ and so by the Cancellation Law $x = 0$ or $y = 0$.

10 Invariant Subgroups

Invariant Subgroups as also known as Normal Subgroups.

Consider a subgroup H of G. For $g, g' \in G$ let $x = gH$ and $y = g'H$ be the corresponding cosets of g and g'. For H to be an Invariant Subgroup of G we require that xy to be in the same coset as $gg'H$ or

$$(gH)(g'H) \in gg'H.$$

To obtain the basic relation defining Invariant Subgroups, premultiply both sides by g^{-1}:

$$\begin{aligned} g^{-1}(gH)(g'H) &\in g^{-1}gg'H, \\ H(g'H) &\in g'H. \end{aligned}$$

In the right most H choose the neutral element as e

$$\begin{aligned} Hg'e &\in g'H \\ Hg' &\in g'H \end{aligned}$$

If $Hg' \in g'H$ for any $g' \in G$ we also must have that (changing g' to g) that $Hg \in gH$. Since $g^{-1} \in G$ we also have

$$Hg^{-1} \in g^{-1}H.$$

and therefore

$$\begin{aligned} (gH)g^{-1}g &\in gg^{-1}Hg, \\ gH &\in Hg, \end{aligned}$$

Therefore we have $gH \in Hg$ and $Hg \in gH$ and so

$$gH = Hg,$$

leading up to the following definition of an Invariant or Normal Subgroup:

Definition H is an invariant subgroup of G if it is a subgroup and if the left and right cosets of H in G coincide; i.e., if

$$gH = Hg \quad \forall g \in G.$$

Invariant subgroups are often called normal subgroups or self-conjugate subgroups.

If the left coset gH is also a right coset, then it contains $g(= eg)$ and so must be the right coset Hg. Hence if every left coset is a right coset (or vice versa) H is invariant.

Let us show that this condition is sufficient.

Theorem 10.1 *If H is invariant in G then $(gH)(g'H) = gg'H$ for all $g, g' \in G$.*

Proof

$$\begin{aligned} gHg'H &= g(Hg')H = g(g'H)H \text{ since } Hg' = g'H \text{ as H is invariant,} \\ &= gg'H^2 = gg'H \text{ since } H^2 = H \text{ as } H \text{ is a subgroup.} \end{aligned}$$

Note we have already shown that it is sufficient to prove that

$$(gH)(g'H) \subseteq gg'H,$$

in order to show that H is an Invariant Subgroup.

Theorem 10.2 *A subgroup H is an invariant subgroup of G is and only if $g^{-1}Hg = H$ for all $g \in G$.*

11 Conjugacy

Definition The elements x and $g^{-1}xg$, where x and g are elements of a group G, are called conjugate elements in G : in other words x and y are conjugate if and only if there exists and element $g \in G$ such that $y = g^{-1}xg$.

Theorem 11.1 *The relation of conjugacy between elements of any group G is an equivalence relations.*

Reflexive. Since $x = e^{-1}x$, x is conjugate to itself.

Symmetric. If $y = g^{-1}xg$ we have $x = gyg^{-1} = (g^{-1})^{-1}yg^{-1}$.

Transitive. If $y = g^{-1}xg$ and $z = h^{-1}yh$ we have

$$z = h^{-1}g^{-1}xgh = (gh)^{-1}x(gh).$$

It follows that the relation divides the elements of G into mutually exclusive classes; these are called *conjugacy classes in G*.

Theorem 11.2 *If H is a subgroup of G then $g^{-1}Hg$ is a subgroup for any element $g \in G$.*

Proof For

$$(g^{-1}Hg)^2 = (g^{-1}Hgg^{-1}Hg) = (g^{-1}HHg) = (g^{-1}Hg) \text{ since } H^2 = H$$

as H is a subgroup. Also

$$(g^{-1}Hg)^{-1} = g^{-1}H^{-1}g = g^{-1}Hg \text{ since } H^{-1} = H.$$

We say that H and $g^{-1}Hg$ are *conjugate subgroups* in G.

Theorem 11.3 *The relation of conjugacy between subgroups of G is an equivalence relation.*

Theorem 11.4 *Normal subgroups of G are composed of all elements of g which are common to the set of conjugate subgroups of G.*

Proof Suppose the set of conjugate subgroups of G are

$$H_1, H_2, \ldots H_i.$$

The elements which are common to these subgroups form a group, since with any two of them contained in H_i their product is also contained in H_i. Also, the transformations of H_1 by elements in G are the same conjugate subgroups

$$H_1, g_2^{-1}H_1g_2, \ldots, g_i^{-1}H_1g_i,$$

the group of common elements, which is part of any one of them as it is of H_1, is identical with its transforms and hence a normal subgroup of G.

A group of all elements which are common to conjugate subgroups, is called their **greatest common subgroup** and a greatest common subgroup is normal in the group.

Theorem 11.5 *H is an invariant subgroup of G if the conjugacy class of subgroups of G which contains H consists of just H itself: i.e., if the only subgroup conjugate to H is H.*

12 Quotient groups

Theorem 12.1 *If H is an invariant subgroup of G, the cosets of H in G form a group under the group product given by*

$$(gH)(g'H) = gg'H.$$

The product is well defined and unique, and is also a coset.

It is associative, since

$$((gH)(g'H))(g''H) = gg'g''H = (gH)((g'H)(g''H)).$$

The coset $H = (eH)$ is a unity since

$$(gH)(eH) = geH = gH \text{ and } (eH)(gH) = egH = gH.$$

The coset gH has an inverse $g^{-1}H$, since

$$(gH)(g^{-1}H) = gg^{-1}H = eH = H \text{ and } (g^{-1}H)(gH) = H$$

similarly.
The quotient group formed by the cosets of H and G is called the *Quotient Group of H in G*, or sometimes the *Factor Group of G relative to H*.

13 Quotient Group Associated with a Homomorphism

As we have already hinted, the subjects of invariant subgroups and homomorphisms of groups are closely connectedl there is in fact a 1-1 correspondence between subgroups of a given group G and the essentially different homomorphism which have G ax the object space, so that the study of either tops is similar to that of the other.

Suppose that H is an invariant subgroup of G, so that the quotient group G/H is defined. Consider the mapping $\theta : G \to G/H$ defined by $g\theta = gH$, the coset containing g, which is of course an element of G/H.

Theorem 13.1 *The mapping $\theta : G \to G/H$ defined by $g\theta = gH$ is a homomorphism.*

Proof

$$\begin{aligned}(gg')\theta &= gg'H = gg'HH \\ &= gHg'H \\ &= (g\theta)(g'\theta)\end{aligned}$$

Corollary 13.2 *The kernel θ is H and the image is G/H, so that θ is an epimorphism.*

The homomorphism θ is called the Natural Homomorphism or the Canonical Homomorphism associated with H. It is essential that H be an invariant subgroup, for otherwise G/H is not defined and we cannot speak of the product of the cosets $(gH)(g'H)$.
Any homomorphism $\theta : G \to G'$ is closely related to the natural homomorphsim associated with an invariant subgroup of G.

Theorem 13.3 *The kernel of $\theta : G \to G'$ is an invariant subgroup of G.*

If h is in the kernel H and g is any element of G, we have

$$\begin{aligned}(g^{-1}hg)\theta &= (g^{-1}\theta)(h\theta)(g\theta) \\ &= (g\theta)^{-1})f(g\theta), \text{ where } f \text{ is the neutral element of } G' \\ &= (g\theta)^{-1}(g\theta) = f\end{aligned}$$

Theorem 13.4 *The mapping defined by $gH \to g\theta$ is a well-defined mapping of G/H onto $G\theta$.*

If g_1 and g_2 are in the same coset, then $g_1\theta = g_2\theta$. If g_1 and g_2 are in the same coset, then $g_2^{-1}g_1 \in H$. Hence $(g_2\theta)^{-1}(g_1\theta) = f$, the neutral element in G', since H is in the kernel of θ. Therefore $g_2\theta = g_1\theta$.
Also any element of $G\theta$ is the image of some $g \in G$ and so the mapping is onto $G\theta$, although not necessarily onto G'.

Theorem 13.5 *The mapping defined by $gH \to g\theta$ is an isomorphism between G/H and $G\theta$.*

We have that

$$\begin{aligned}(g_1H)(g_2H) &= g_1g_2H, \\ (g_1\theta)(g_2\theta) &= (g_1g_2)\theta,\end{aligned}$$

because θ is a homomorphism and $g_1g_2H \to (g_1g_2)\theta$.
Since this mapping is onto $G\theta$, it is also an isomorphism provided it is 1-1, i.e., provided the kernel is the neutral element of G/H. Suppose $gH \to f$. Then $g\theta \to f$ and so $g \in H$, so that $gH = H$, the neutral element of G/H.

Notice that the sets of elements of G which map into the particular elements of $G\theta$ are the cosets of G relative to the kernel H.

We have proved that to any invariant subgroup H of G there is a natural homomorphism from G onto G/H, and conversely that any homomorphism of G is isomorphic to a natural homomorphism associated with its kernel. Therefore there is a 1-1 correspondence between the invariant subgroups and the essential different homomorphisms of G. By essentially different homomorphisms we mean those having different kernels: two homomorphisms having the same kernel are both equivalent to the same natural homomorphism and therefore exhibit the same structure.

The image space G' is largely irrelevant : the subgroup $G\theta$ of G' is isomorphic to G/H but G' may be larger. Also there is no reason why $G\theta$ should be invariant in G'.

14 Relationship between Normal Subgroups and Their Images

Consider a group G that has a normal subgroup N and H and that $H \subseteq N$.

Theorem 14.1 *Because $N \supseteq H$, both N and H are normal subgroups of G. Thus G/N is the homomorphic image of G/H under the natural homomorphsim $\pi : G/H \to G/N$ defined by $(xG)\pi = xN$.*

Proof 1) Define a function $\phi : G \to G/N$ by

$$x\phi = xN.$$

This is the standard canonical projection map. It is a surjective group homomorphism with $ker(\phi) = N$.
2) We want to construct a map $\overline{\phi} : G/H \to G/N$. Define $\overline{\phi}$ by:

$$(xH)\overline{\phi} = x\phi = xN.$$

3) We must show that if $xH = yH$, then $(xH)\overline{\phi} = (yH)\overline{\phi}$.

$$\begin{aligned} xH &= yH \to y^{-1}x \in H. \\ H &\subseteq N \text{ given} \\ y^{-1}x &\in N \\ xN &= N \\ (xH)\overline{\phi} = (yH)\overline{\phi} & \end{aligned}$$

The map is well-defined because $H \subseteq ker(\phi)$.

4) $\overline{\phi}$ is a Homomorphism:

$$(xH)(yH)\overline{\phi} = (xyH)\overline{\phi} = xyN = (xN)(yN) = (xH)\overline{\phi}(yH)\overline{\phi}.$$

5) $\overline{\phi}$ is Surjective: Let $yN \in G/N$. Since $y \in G$, we can form the coset $yH \in G/H$:

$$(yH)\overline{\phi} = yN.$$

Every element in the co-domain has a pre-image

6) Apply the First Isomorphism Theorem By the First Isomorphism Theorem, the domain modulo the kernel is isomorphic to the image

$$(G/H)/ker(\overline{\phi}) \cong Im(\overline{\phi}) = G/N.$$

Since G/N is isomorphic to a quotient group of G/H, it is by definition a homomorphic image of G/H. ∎

Lattice Isomorphism Theorems

Theorem 14.2 (First Isomorphism Theorem:) *Let ϕ be a group homomorphism from G to G'. Then the mapping from $G/Ker(\phi)$ to $G\phi$ given by $gKer(\phi) \to g\phi$ is an isomorphism.*

Example Consider the homomorphism $\phi : \mathbb{Z} \to \mathbb{Z}_4$ such that $x\phi = x \mod 4$. In this case let G be the group of integers (under addition)

$$G = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...\}$$

and

$$G' = \mathbb{Z}_4 = \{0, 1, 2, 3\}.$$

Also

$$Ker(\phi) = \{... - 8, , -4, 0, 4, 8, ...\} = \langle 4 \rangle \text{ (the cyclic subgroup generated by 4)}$$

So $G\phi = \{0, 1, 2, 3\}$. Note that $G\phi$ doesn't have to be the entire group G'. In this example it is the case that $G\phi = G'$. Then

$$G/Ker(\phi) = \{0 + \langle 4 \rangle, 1 + \langle 4 \rangle, 2 + \langle 4 \rangle, 3 + \langle 4 \rangle\} \text{ (four cosets)}$$

Notice that we can create a mapping $\overline{\phi}$ that mapps $G/Ker(\phi)$ into $G\phi$ as follows:

$$\overline{\phi} : G/Ker(\phi) \to g\phi,$$

where

$$\begin{aligned}
\overline{\phi}(0 + \langle 4 \rangle) &= (0)\phi = 0 \\
\overline{\phi}(1 + \langle 4 \rangle) &= (1)\phi = 1 \\
\overline{\phi}(2 + \langle 4 \rangle) &= (2)\phi = 2 \\
\overline{\phi}(3 + \langle 4 \rangle) &= (3)\phi = 3
\end{aligned}$$

Theorem 14.3 (Second Isomorphism Theorem:) *The Second Isomorphism Theorem: If S is a subgroup of G and N is a normal subgroup of G , then the product SN is a subgroup, and the quotient SN/N is isomorphic to $S/(S \cap N$.*

Theorem 14.4 (Third Isomorphism Theorem:) *Normal Subgroups of Quotients: If N and H are normal subgroups of G (where $N \subseteq H$), then H/N is is a normal subgroup of G/N. Furthermore, the second quotient is isomorphic to the primary quotient:*

$$(G/N)/(H/N) \cong G/H.$$

The Third Isomorphism Theorem is analogous to fractions:

$$\frac{G/N}{H/N} \cong \frac{G}{H}.$$

Theorem 14.5 (Fourth Isomorphism Theorem:) *There is a direct, structure-preserving correspondence between the subgroups of a quotient group G/N and the subgroups of G that contain N.*

Theorem 14.6 *The mapping $\phi : \mathbb{Z} \to \mathbb{Z}_n$ defined by $m\phi = m \mod n$ is a homomorphism with kernel $\langle n \rangle$. By the First Isomorphism Theorem, $\mathbb{Z}/\langle n \rangle \cong \mathbb{Z}_n$.*

If ϕ is a homomorphism from a finite group G to G', then $\|G\phi\|$ divides $\|G\}$ and $\|G'\|$. This follows from the First Isomorphism Theorem and the fact that $G\phi$ is a subgroup of G' and Lagrange's Theorem.

Lemma 14.7 *The inner automorphism θ_x given by $g \to x^{-1}gx$ is the identity automorphism if and only if $x \in C$ where C is the center.*

If $x \in C$, $x^{-1}gx = x^{-1}xg = g$ and so $\theta_x = I$, the identity.
If $\theta_x = I, x^{-1}gx = g\forall g$ and so $x \in C$.

Lemma 14.8 *The mapping $x \to \theta_x$ of G into the group of automorphisms of G is a homomorphism*

We already have that $\theta_{xy} = \theta_x\theta_y$ and so the result follows.
By Lemma(14.7) the kernel of this homomorphism is the center C. The image is the subgroup of inner automorphisms. Therefore we obtain

Theorem 14.9 *The group of inner automorphisms of a group $G \cong G/C$, where C is the center of G.*

15 Fundamental Theorem of Algebra

Axiom I - Axiom of inequality for $\mathbb{R}$. Among the real numbers, $\mathbb{R}$, there is a set called the positive numbers which satisfies the conditions: (i) for any number $a \in \mathbb{R}$ exactly one of the three alternatives holds; a is positive, $a = 0$, or $-a$ is positive; (ii) any finite sum or produce of positive numbers is positive.

Axiom C - Axiom of continuity. Suppose that an infinite sequence $x_1, x_2, \ldots, x_n \ldots$ is such that $x_{n+1} \geq x_n$ for all n, and there is a number $M \in \mathbb{R}$ such that $x_n \leq M$ for all n. Then there is a number $L \leq M$ such that $x_n \to L$ as $n \to \infty$ and $x_n \leq L$ for all n.

Theorem 15.1 *(Nested Intervals Therorem): Suppose that*

$$I_n = \{x : a_n \leq x \leq b_n\}, \quad n = 1, 2, ...,$$

is a sequence of closed intervals such that $I_{n+1} \subset I_n$, for each n. If $\lim_{n\to\infty}(b_n - a_n) = 0$, then there is one and only one number x_0 which is in every I_n.

Proof By hypothesis, we have $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$. Since $a_n < b_n$ for every n, the sequence $\{a_n\}$ is nondecreasing and bounded from above by b_1. Similarly, the sequence b_n is non-increasing and bounded from below by a_1. Using Axiom C, we conclude that there are numbers x_0 and x_0' such that $a_n \to x_0, a_n \leq x_0$, and $b_n \to x_0', b_n \geq x_0'$ as $n \to \infty$. Using the fact that $b_n - a_n \to 0$ as $n \to \infty$, we find that $x_0 = x_0'$ and

$$a_n \leq x_0 \leq b_n$$

for every n. Thus x_0 is in every I_n.
To show that x_0 is the only number in every I_n, suppose there is another number x_1 also in every I_n. Define $\epsilon = |x_1 - x_0|$. Now, since $a_n \to x_0$ and $b_n \to x_0$, there is an integer N_1, such that $a_{N_2} > x_0 - \epsilon$; also there is an integer N_2 such that $b_{N_2} < x_0 + \epsilon$. These inequalities simply that I_n cannot contain x_1 for any n beyond N_1 and N_2. Thus x_1 is not in every I_n. ∎

Theorem 15.2 *Suppose f is continuous on $[a, b], x_n \in [a, b]$ for each n, and $x_n \to x_0$. Then $x_0 \in [a, b]$ and $f(x_n) \to f(x_0)$.*

Proof Since $x_n \geq a$ for each n, it follows from the theorem on the limit of inequalities for sequences that $x_0 \geq a$. Likewise $x_0 \leq b$ so $x_0 \in [a, b]$. If $a < x_0 < b$, then $f(x_n) \to f(x_0)$ on account of the composite function theorem. The same conclusion holds if $x_0 = a$ or b.

Theorem 15.3 *(Intermediate-value theorem): Suppose f is continuous on $[a, b], c \in \mathbb{R}, f(a) < c$, and $f(b) > c$. Then there is at least one number x_0 on $[a, b]$ such that $f(x_0) = c$.*

There may be more than one such point!

Proof Define $a_1 = 1$ and $b_1 = b$. Then observe that $f((a_1 + b_1)/2)$ is either equal to c, greater than c, or less than c. If it equals c, then choose $x_0 = (a_1 + b_1)/2$ and the result is proved. If $f((a_1 + b_1)/2) > c$, then define $a_2 = a_1$ and $b_2 = (a_1 + b_1)/2$. If $f((a_1 + b_1)/2 < c$, then define $a_2 = (a_1 + b_1)/2$ and $b_2 = b_1$.
In each of the last two cases we have $f(a_2) < c$ and $f(b_2) > c$. Again compute $f((a_2 + b_2)/2)$. If this value equals c, the result is proved. If $f((a_2 + b_2)/2) > c$, set $a_3 = a_2$ and $b_3 = (a_2 + b_2)/2$. If $f((a_2 + b_2)/2 < c$, then define $a_3 = (a_2 + b_2)/2$ and $b_3 = b_2$.
Continuing in this way, we either find a solution in a finite number of steps or we find a sequence $\{[a_n, b_n]\}$ of closed intervals each of which is one of the two halves of the preceding one, and for which we have

$$b_n - a_n = 2^{1-n}(b_1 - a_1),$$

$$f(a_n) < c, \quad f(b_n) > c, \text{ for each } n,$$

From the Nested Intervals Theorem it follows that there is a unique point x_0 in all these intervals and

$$\lim_{n \to \infty} a_n = x_0 \text{ and } \lim_{n \to \infty} b_n = x_0.$$

From the theorem of continuous function on a sequence of points in a closed interval, we conclude that

$$f(a_n) \to f(x_0) \text{ and } f(b_n) \to f(x_0).$$

From the inequalities about and the limit of inequalities it follows that $f(x_0) \leq c$ and $f(x_0) \geq c$ so that $f(x_0) = c$.

Theorem 15.4 (Extreme Value Theorem for real-valued functions of two real variables:) *Let D be the* **closed disk** *or radius $r > 0$ and centered at the origin, i.e,*

$$D = \{(x_1, x_2) \in \mathbb{R}^2 | x_1^2 + x_2^2 \leq r\}.$$

Let $f : D \to \mathbb{R}$ be a continuous function on the closed disk $D \subset \mathbb{R}^2$. Then f is bounded and attains its minimum and maximum values on D. In other words, there exists points $x_m, x_M \in D$ such that

$$f(x_m) \leq f(x) \leq f(x_M)$$

for every possible choice of point $x \in D$.

Definition A polynomial over a field F is a *rational* and *integral* expression of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0, \tag{1}$$

where the coefficients $a_i \in F, i = 1, 2, \ldots, n$ and *rational* means free from fractional powers and *integral* means never in the denominator of a fraction. A polynomial function is a function which can be written in the form of Eq. (1).

Theorem 15.5 (Eisenstein's Irreducibility Criterion) *For a given prime p, let $a(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots a_0$ be a polynomial with integral coefficients, such that $a_n \neq 0(\mod p)$, $\quad a_{n-1} = a_{n-2} = a_0 = 0(\mod p), a_0 \neq 0(\mod p^2)$. Then $a(x)$ is irreducible over the field of rational.*

Proof 1 Suppose

$$a(x) = gh,$$

where

$$\begin{aligned} g &= g_0 x^h + g_1 x^{h-1} + \ldots + g_{k-1}x + g_k \\ h &= h_0 x^l + h_1 x^{l-1} + \ldots + h_{l-1}x + h_l \end{aligned}$$

Then g and h have integral coefficients since f has. Comparing coefficients, we set

$$\begin{aligned} a_0 &= g_k h_l \\ a_1 &= g_k h_{l-1} + g_{k-1}h_l \\ a_2 &= g_k h_{l-2} + g_{k-1}h_{l-1} + \ldots + g_{k-2}h_l \\ \ldots & \quad \ldots \\ a_l &= g_k h_{l-k} + g_{k-1}h_{l-k+1} + \ldots + g_1 h_{l-1} + h_l \\ \ldots & \quad \ldots, \end{aligned}$$

where the h_i with possible negative subscripts are 0 and where

$$h_0 = 1.$$

if now every a_i, other than a_0 is divisible by p and a_n not by p^2, we conclude from the first equation that either g_k or h_l is divisible by p and the other is not. But when g_k is divisible by p, we conclude from the following equations that $g_{k-1}, g_{k-2}, \ldots, g_1$ are divisible by p, and from the last equation that h_l is also divisible by p which is a contradiction from the assumption that a_n is not divisible by p^2.

Proof 2 In any possible factorization $(n = m + k)$

$$a(x) = bc = (b_n x^n + b_{n-1}x^{n-1} + \ldots + b_1 x + b_0)(c_k x^k + c_{k-1}x^{k-1} + \ldots + c_1 x + c_0)$$

we may assume that both factors have integral coefficients. Since $a_0 = b_0 c_0$ the hypothesis $a_0 \not\equiv 0 \pmod{p^2}$ means that not both b_0 and c_0 are divisible by p. Suppose $b_0 \not\equiv 0 \pmod p$, while $c_0 \equiv 0 \pmod p$. But $b_m c_k = a_n \not\equiv 0 \pmod p$, so $c_k \not\equiv 0 \pmod p$. Pick the smallest $r \leq k$ for which $c_r \not\equiv 0 \pmod p$, with $c_{r-1} \equiv \ldots \equiv c_0 \equiv 0 \pmod p$. Then

$$a_r = b_0 c_r + b_1 c_{r-1} + \ldots + b_r c_0 = b_0 c_r \pmod p.$$

But $\boldsymbol{b}_0 \not\equiv 0 \pmod p$ and $c_r \not\equiv 0 \pmod p$ give $a_r \not\equiv 0 \pmod p$, for p is a prime. By the hypothesis, the only coefficient a_r for which this is possible is a_n, therefore $r = n$: the degree of the second of the proposed factors must be n (the same as degree of $a(x)$, so that $a(x)$ is irreducible. ∎

This criterion can be applied to the cyclotomic polynomial

$$\phi(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \ldots + x + 1. \tag{2}$$

This is the polynomial whose roots are the primitive p-th roots of unity. The Eisenstein criterion does not apply to Eq. (2) as it stands, but a simple change of variables $y = x - 1$ does the trick, for the binomial expansion gives

$$\frac{x^p - 1}{x - 1} = \frac{(y+1)^p - 1}{y} = y^{p-1} + py^{p-2} + \frac{p(p-1)}{1 \cdot 2} y^{p-3} + \ldots + p.$$

The binomial coefficients which appear on the right are all integers divisible by the prime p, for p occurs in each numerator as a factor and can never be cancelled out by the smaller integers in the denominator. The polynomial in y satisfies the hypothesis of the Eisenstein criterion and hence is irreducible. Therefore the original cyclotomic polynomial, Eq. (2), $\phi(x)$ is irreducible.

Theorem 15.6 (Euler-Gauss: Fundamental Theorem of Algebra) *Every polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$ of positive degree with complex coefficients has a complex root.*

We can regard $(x, y) \to |f(x+iy)|$ as a function $\mathbb{R}^2 \to \mathbb{R}$. We denote this function by $|f(\cdot)|$ or $|f|$. Since polynomials and the square root function are continuous, then $|f|$ is also continuous.

Lemma 15.7 *Let $f : \mathbb{C} \to \mathbb{C}$ be any polynomial function. Then there exists a point $z_0 \in \mathbb{C}$ where the function $|f|$ attains a minimum value in $\mathbb{R}$.*

Proof If f is a constant polynomial function, then the statement of Lemma (15.7) is trivial, so we may take $z_0 = 0$. If f is not constant, then the degree of the polynomial defining f is at least one. In this case, we can denote f expliciitly as Eq. (1) with $a_n \neq 0$. Now assume that $z \neq 0$, and set $A = max\{|a_0|, |a_1|, \ldots, a_{n-1}|\}$. We can obtain a lower bound for $|f(z)$ as follows:

$$\begin{aligned} |f(z)| &= |a_n||z^n||1 + \frac{a_{n-1}}{a_n z} + \ldots + \frac{a_0}{a_n z^n}| \\ &\geq |a_n||z^n|| \left(1 - \frac{A}{|a_n|} \sum_{k=1}^{\infty} \frac{1}{|z|^k}\right) = |a_n||z^n|| \left(1 - \frac{A}{|a_n|} \frac{1}{|z| - 1}\right) \end{aligned}$$

For all $z \in \mathbb{C}$ such that $|z|| \geq 2$, we can further simplify this expression to

$$|f(z)| \geq |a_n||z^n| \left(1 - \frac{2A}{|a_n||z||}\right).$$

It follows from this inequality that there is an $r > 0$ such that $|f(z)| > |f(0)|$, for all $z \in \mathbb{C}$ satisfying $|z| > r$. Let $D \subset \mathbb{R}^2$ be the disk of radius r centered at 0, and define a function $g : D \to \mathbb{R}$, by

$$g(x, y) = |f(x + iy)|.$$

Since g is continuous, we can apply Thm (15.4) in order to obtain a point $(x_0, y_0) \in D$ such that g attains its minimum at (x_0, y_0). By the choice of r we have that for $z \in C\ D, |f(z)| > |g(0,0)| \geq |g(x_0, y_0)|$. Therefore $|f|$ attains its minimum at $z = x_0 + iy_0$.

Proof of Thm (15.6): We rely on the fact that the function $|f(x)|$ attains its minimum value by Lemma (15.7). Let $z_0 \in \mathbb{C}$ be a point where the minimum is attained. We will show that if $f(z_0) \neq 0$, then z_0 is not a minimum value, thus proving that the minimum value of $|f(z)|$ is zero and therefore $f(z_0) = 0$.

If $f(z_0) \neq 0$, then we can define a new function $g : \mathbb{C} \to \mathbb{C}$ by setting

$$g(z) = \frac{f(z + z_0)}{f(z_0)}, \text{ for all } z \in \mathbb{C}.$$

Not that g is a polynomial of degree n, and that the minimum of $|f|$ is attained at z_0 if and only if the minimum of $|g|$ is attained at $z = 0$. Moreover, it is clear that $g(0) = 1$. Therefore g is a polynomial of the form

$$g(z) = b_n z^n + \ldots + b_k z^k + 1,$$

with $n \geq 1$ and $b_k \neq 0$ for some $1 \leq k \leq n$. Let $b_k = |b_k| e^{i\theta}$, and consider z of the form

$$z = r|b_k|^{-1/k} \, e^{i(\pi - \theta)/k}, \tag{3}$$

with $r > 0$. For z of this form we have

$$g(z) = 1 - r^k + r^{k+1} h(r),$$

with h a polynomial. Then for $r < 1$, we have by the triangle inequality that

$$|g(z)| \leq 1 - r^k + r^{k+1} |h(r)|.$$

For $r > 0$ sufficiently small we have $r|h(r)| < 1$, by the continuity of the function $rh(r)$ and the fact that it vanishes at $r = 0$. Hence we find that

$$|g(z)| \leq 1 - r^k (1 - r|h(r)|) < 1,$$

for some z having the form given in Eq. (3) with $r \in (0, r_0)$ and $r > 0$ sufficiently small. But then the minimum of $|g| : \mathbb{C} \to \mathbb{R}$ cannot possibly be equal to 1. ∎

16 Elementary Symmetric Functions

If $a_n = 1$ we say the polynomial is in *monic* form. It will often be convenient to adopt this convention; and in such cases we shall write the polynomial as

$$f(x) = x^n + p_1 x^{n-1} + p_2 x^{n-2} + \ldots + p_{n-1} x + p_n.$$

The roots $\{\alpha_1, \alpha_2, \ldots \alpha_n\}$ of a polynomial are the values of x such that $f(x = \alpha_i) = 0$. The roots are in general complex numbers. According to a theorem of Viète, the relationship between the roots and the coefficients for a monic polynomial is given by the

$$\begin{aligned} f(x) &= x^n + p_1 x^{n-1} + \ldots + p_{n-1} x + p_n \\ &= (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \ldots (x - \alpha_n). \end{aligned}$$

Expanding the product on the right-hand side we find the explicit relationships:

$$
\begin{aligned}
p_1 &= -(\alpha_1 + \alpha_2 + \ldots + \alpha_n) \\
p_2 &= (\alpha_1\alpha_2 + \alpha_1\alpha_3 + \ldots + \alpha_2\alpha_3 + \ldots + \alpha_{n-1}\alpha_n) \\
p_3 &= -(\alpha_1\alpha_2\alpha_3 + \alpha_1\alpha_2\alpha_4 + \ldots + \alpha_2\alpha_3\alpha_4 + \ldots + \alpha_{n-2}\alpha_{n-1}\alpha_n) \\
&\vdots \\
p_n &= -(1)^n(\alpha_1\alpha_2\alpha_3\alpha_4 \ldots \alpha_{n-1}\alpha_n)
\end{aligned}
$$

These relationships are encapsulated in the following theorem

Theorem 16.1 *In every algebraic equation, the coefficient of whose highest term is unity, the coefficient p_1 of the second term with its sign changed is equal to the sum of the roots.*

The coefficient p_2 of the third terms is equal to the sum of the products of the roots taken two by two.

The coefficient of the fourth term with its sign changed is equal to the sum of the products of the roots taken three by three; and so on, the signs of the coefficients being taken alternately negative and positive, and the number of roots multiplied together in each term of the corresponding function of the roots increasing by unity till finally that function is reached with consists of the produces of the n roots.

Definition The Elementary Symmetric functions $c_1, c_2, c_3, \ldots c_n$ are defined as functions of n indeterminants $x_1, x_2, x_3, \ldots, x_n$ as follows

$$
\begin{array}{lll}
s(x_1) & = x_1 + x_2 + \ldots + x_n & = c_1 \\
s(x_1x_2) & = x_1x_2 + x_1x_3 + \ldots + x_2x_3 + \ldots + x_{n-1}x_n & = c_2 \\
s(x_1x_2, x_3) & = x_1x_2x_3 + x_1x_2x_4 + \ldots + x_2x_3x_4 + \ldots + x_{n-2}x_{n-1}x_n & = c_3 \\
\cdots & \cdots\cdots\cdots & \cdots \\
s(x_1x_2 \ldots x_n) & = x_1x_2x_3x_4 \ldots x_{n-1}x_n & = c_n
\end{array} \tag{4}
$$

Theorem 16.2 *The sums of the similare powers of the roots of an equation can be expressed in terms of the coefficients (the elementary symmetric functions).*

Proof Consider the polynomial

$$
\begin{aligned}
f(x) &= x^n + p_1x^{n-1} + p_2x^{n-2} \ldots + p_{n-1}x + p_n, \\
&= (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \ldots (x - \alpha_n) = 0.
\end{aligned}
$$

Define the following terms for the sums of the powers of the roots:

$$
s_i = \sum_{j=1}^{n} \alpha_j^i, \quad i = 0, 1, 2, ...
$$

Given that $f(x)$ is a polynomial we can write the derivative of $f(x)$ as

$$
\begin{aligned}
f'(x) &= \frac{f(x)}{x - \alpha_1} + \frac{f(x)}{x - \alpha_1} + \ldots + \frac{f(x)}{x - \alpha_1} \\
&= nx^{n-1} + (n-1)p_1x^{n-2} + (n-2)x^{n-3} + \ldots + 2p_{n-2}x + p_{n-1}
\end{aligned}
$$

Using the method of Synthetic Division to divide $f(x)$ by the monomial $(x - a)$ we write out the sequence of computations as follows;

$$\begin{array}{rr|r|r|r}
\frac{f(x)}{x-a} & = x^{n-1} + a & x^{n-2} + a^2 & x^{n-3} + a^3 & x^{n-4} + \ldots + a^{n-2} \\
 & +p_1 & +ap_1 & +a^2p_1 & +a^{n-2}p_1 \\
 & & +p_2 & +ap_2 & +a^{n-3}p_2 \\
 & & & +p_3 & + \ldots \\
 & & & & +ap_{n-2} \\
 & & & & +p_{n-1}
\end{array}$$

By replacing a by each of the quantities $a_1, a_2, ...a_n$ in succession and adding and using $s_p = \sum a^p$ we obtain the following expression for the derivative of $f(x)$:

$$\begin{array}{rr|r|r|r}
f'(x) & = x^{n-1} + s_1 & x^{n-2} + s_2 & x^{n-3} + s_3 & x^{n-4} + \ldots + s_{n-2} \\
 & +np_1 & +s_1p_1 & +s_2p_1 & +s_{n-2}p_1 \\
 & & +np_2 & +sp_2 & +s_{n-3}p_2 \\
 & & & +np_3 & + \ldots \\
 & & & & +sp_{n-2} \\
 & & & & +np_{n-1}
\end{array}$$

Therefore we obtain the following relationships between the s_i's and the coefficients p_i:

$$\begin{array}{rcl}
s_1 + p & = & 0 \\
s_2 + p_1 s_1 + 2p_2 & = & 0 \\
s_3 + p_1 s_2 + p_2 s_1 + 3p_3 & = & 0 \\
s_4 + p_1 s_3 + p_2 s_3 + p_3 s_1 + 4p_4 & = & 0 \\
\ldots\ldots\ldots\ldots & & \\
s_{n-1} + p_1 s_{n-2} + p_2 s_{n-3} + \ldots + p_{n-2} s_1 + (n-1)p_{n-1} & = & 0
\end{array}$$

We now proceed to extend our results to the sums of all positive powers of the roots, $s_n, s_{n+1}, \ldots, s_m$ by multiplying the polynomial by x^{m-n} and using the $s_0 = n$, we obtain

$$x^{m-n} f(x) = x^m + p_1^{m-1} + p_2 x^{m-2} + \ldots + p_n x^{m-n},$$

By replacing x by the roots $a_1, a_2, \ldots, a_n$ in succession and adding we have,

$$s_m + p_1 s_{m-1} + p_2 s_{m-2} + \ldots + p_n s_{m-n} = 0.$$

Letting m have the values $m = n, n+1, n+2, \ldots$ successively and noting that $s_0 = n$, we obtain the following equations

$$\begin{array}{rcl}
s_n + p_1 s_{n-1} + p_2 s_{n-2} + \ldots + np_n & = & 0 \\
s_{n+1} + p_1 s_n + p_2 s_{n-1} + \ldots + p_n s_1 & = & 0 \\
s_{n+2} + p_1 s_{n+1} + p_2 s_n + \ldots + p_n s_2 & = & 0 \\
\ldots\ldots\ldots & & 0
\end{array}$$

By transforming the equation $f(x) = 0$ into one whose roots are the reciprocals of $a_1, a_2, a_3, \ldots, a_n$, and applying the above formulas we can also express all the negative powers of the roots.

Theorem 16.3 *Every rational symmetric function of $x_1, x_2, \ldots, x_n$ is rationally expressible in terms of the elementary symmetric functions.*

Proof It is sufficient to prove this theorem for integral functions only, since fractional symmetric functions can be reduced to a single fraction whose numerator and denominator are both integral symmetric functions. Every integral function of $a_1, a_2, a_3, \ldots a_n$ is the sum of a number of terms of the form $N a_1^p a_2^q a_3^r \ldots$, where N is a numerical constant; and if this function be symmetrical we can write it under the form $N \sum a_1^p a_2^q a_3^r \ldots$, all the terms being of the same type. Therefore if we can prove that this quantity be expressed rationally in terms of the coefficients, the theorem will be demonstrated.

Let us begin with the symmetric function

$$\sum a_1^p a_2^q = s_p s_q - s_{p+q}. \tag{5}$$

Which can be proven by multiplying s_p and s_q, where

$$\begin{aligned} s_p &= a_1^p + a_2^p + a_3^p + \ldots + a_n^p, \\ s_q &= a_1^q + a_2^q + a_3^q + \ldots + a_n^q, \end{aligned}$$

Therefore

$$\begin{aligned} s_p s_q &= a_1^{p+q} + a_2^{p+q} + a_3^{p+q} + \ldots + a_n^{p+q} + a_1^p a + 2^q + a_1^p a + 3^q + \ldots, \text{ or} \\ s_p s_q &= s_{p+q} + \sum a_1^q a_2^q, \end{aligned}$$

and finally

$$\sum a_1^p a_2^p = s_p s_q - s_{p+q}.$$

We proceed now to prove a similar expression for the triple function,

$$\sum a_1^p a_2^q a_3^r = s_p s_q s_r - s_{q+r} s_p - s_{r+p} s_q - s_{p+q} s_r + 2 s_{p+q+r}. \tag{6}$$

Multiplying together $\sum a_1^p a_2^q$ and s_r, where

$$\begin{aligned} \sum a_1^p a_2^q &= a_1^p a_2^q + a_1^q a_q^p + + a_1^p a_3^q + \ldots, \\ s_r &= a_1^r + a_2^r + a_3^r + \ldots + a_n^r \end{aligned}$$

we obtain an expression consisting of three different parts of the forms $\sum a_1^{p+r} a_2^q$, $\sum a_1^{q+r} a_2^p$, and $\sum a_1^p a_2^q a_3^r$, a formula connecting double and triple symmetric functions. However, by Eq(5):

$$\begin{aligned} \sum a_1^{p+r} a_2^q &= s_{p+r} s_q - s_{p+q+r}, \\ \sum a_1^{q+r} a_2^p &= s_{q+r} s_p - s_{p+q+r}, \\ \sum a_1^p a_2^q &= s_p s_q - s_{p+q}, \end{aligned}$$

Substituting these values we find the triple function expressed as in Eq(6).

In the same manner the quadriple function $\sum a_1^p a_2^q a_3^r a_4^s$, can be made to depend on the triple function $\sum a_1^p a_2^q a_3^r$, and ultimately on $s_1, s_2, s_3, \ldots$. Finally every rational symmetric function of the roots may be expressed in terms of the elementary symmetric functions (the coefficients of a polynomial).

The formulas Eq.(5) and Eq.(6) require modification when any of the exponents become equal.

Thus if $p = q$, $a_1^p a_2^q = a_2^p a_1^q$, and the terms in Eq.(5) become equal to two and two; therefore $\sum a_1^p a_2^q = 2 \sum a_1^p a_2^p$ and

$$\sum a_1^p a_2^p = \frac{1}{2}(s_p^2 - s_{2p}).$$

Similarly, if $p = q = r$, the six terms obtained by interchanging the roots in $a_1^p a_2^q a_3^r$ become equal and therefore

$$\sum a_1^p a_2^p a_3^p = \frac{1}{2 \cdot 3}\left(s_p^3 - 3 s_p s_{2p} + 2 s_{3p}\right).$$

And in general, if m exponents become equal, each term is repeated $m!$ times.

Theorem 16.4 (Theorems related to Symmetric Functions) *The sum of the exponents of all the roots in any term of any symmetric function of the roots is equal to the sum of the suffices in each term of the corresponding value in terms of the coefficients.*

The sum of the exponents is of course the same for every term of the symmetric function, and may be called the degree in all the roots of that function. In other words the suffix for each coefficient is equal to the degree in the roots of the corresponding function of the roots; hence in any product of any powers of the coefficients the sum of the suffices must equal to the degree in all the roots of the corresponding function of the roots.

Theorem 16.5 *When an equation is written with binomial coefficients, the expression in terms of the coefficients for any symmetric function of the roots, which is a function of their differences only, is such that the algebraic sum of the numerical factors of all the terms in it is equal to zero.*

Theorem 16.6 (Division Algorithm for Polynomials) *If F is any field, and $a(x) \neq 0$ and $b(x)$ are any two polynomials over F, then we can find polynomials $q(x)$ and $r(x)$ such that*

$$b(x) = q(x)a(x) + r(x),$$

where $r(x)$ is either zero or has a degree less than that of $a(x)$.

Theorem 16.7 *In $F[x]$, any two polynomials a and b have a greatest common divisor (GCD) d satisfying (i) $d|a$ and $d|b$, (ii) $c|a$ and $c|b$ imply that $c|d$. Moreover, (iii) d is a linear combination $d = sa + tb$ of a and b.*

The procedure for determining the GCD of $a(x)$ and $b(x)$ is given in Algorithm (1):

17 Synthetic Division

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$ and consider dividing this polynomial by a linear factor $(x - h)$. This division can certainly be performed by long division, but a more compact method is to use Synthetic Division which is outlined below. Let the quotient when

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$$

Algorithm 1 Polynomial Euclidean Algorithm

```
procedure PolyGCD(a(x), b(x))
    while b(x) ≠ 0 do
        r(x) ← a(x) mod b(x)
        a(x) ← b(x)
        b(x) ← r(x)
    end while
    return a(x)
end procedure
```

is divided by $(x-h)$ be given by

$$Q(x) = b_{n-1}x^{n-1} + b_{n-2}x^{n-2} + \ldots + b_1 x + b_0.$$

We then can represent $f(x)$ with the following identity:

$$f(x) = (x-h)Q(x) + R,$$

with quotient $Q(x)$ and remainder R. The right-hand side of this identity can be grouped according to terms which multiply the same power of x as follows:

$$\begin{array}{r|r|r|r|c|r|r} b_{n-1}x^n & b_{n-2}x^{n-1} & b_{n-3}x^{n-2} & b_{n-4}x^{n-3} & \ldots & b_0 x & R \\ & -hb_{n-1}x^{n-1} & -hb_{n-2}x^{n-2} & -hb_{n-3}x^{n-3} & \ldots & -hb_1 x & -hb_0 \end{array}$$

Equating the coefficients of x on both sides of the we get the following recursive relationships:

$$\begin{array}{rcl} b_{n-1} & = & a_n \\ b_{n-2} & = & hb_{n-1} + a_{n-1} \\ b_{n-3} & = & hb_{n-2} + a_{n-2} \\ \ldots & = & \ldots \\ b_1 & = & hb_2 + a_2 \\ b_0 & = & hb_1 + a_1 \\ R & = & hb_0 + a_0 \end{array}$$

These recursive relations can be represented in the following table (known as Synthetic Division):

$$\begin{array}{c|cccccc} h & a_n & a_{n-1} & \ldots & a_2 & a_1 & a_0 \\ & & hb_{n-1} & \ldots & hb_2 & hb_1 & hb_0 \\ \hline & b_{n-1} & b_{n-2} & \ldots & b_1 & b_0 & R \end{array}$$

As an example let us use Synthetic division to perform the following division:

$$\frac{2x^3 - 7x^2 + 5}{x-3}$$

Synthetic division give us $\begin{array}{c|rrrr} & 2 & -7 & 0 & 5 \\ 3 & & 6 & -3 & -9 \\ \hline & 2 & -1 & -3 & -4 \end{array}$

We can read the terms in the Quotient and Remainder from the terms below the horizontal line in the above.

$$\begin{aligned}2x^3 - 7x^2 + 5 &= (x-3)(2x^2 - x - 3) - 4 \text{ with quotient } Q(x) \text{ and remainder } R \text{ given by:}\\ Q(x) &= 2x^2 - x - 3\\ R &= -4\end{aligned}$$

18 Rational Roots

Theorem 18.1 *If the equation with integral coefficients*

$$a_n x^n + a_{n-1}x^{n-1} + \ldots + a_0 = 0 \tag{7}$$

has a rational root r/s, where r and s are integers without a common divisor other than unity, then r is a divisor of a_0 and s is a divisor of a_n.

Proof If x in Eq. (7) is replaced by r/s and the denominator is cleared of factors of s, then we obtain the following

$$a_n r^n + a_{n-1}r^{n-1}s + a_{n-2}r^{n-2}s^2 + \ldots + a_1 r s^{n-1} = -a_0 s^n.$$

And since r and s are relatively prime, it follows that r is a divisor or a_0, as was to be proved.

Corollary 18.2 *if the coefficients in the equation*

$$x^n + p_1 x^{n-1} + p_2 x^{n-2} + \ldots + p_n = 0$$

are integers, then every rational root of the equation is an integer and a divisor of the constant term p_n.

Proof This corollary follows since s can have only the values ± 1 if $a_n = 1$.

19 Cyclotomic Polynomials

Definition The cyclotomic polynomial $\phi(x)$ for non-trivial p-th roots of unity (other than $x = 1$) is given by

$$\phi(x) = \frac{(x^p - 1)}{x - 1} = x^{p-1} + x^{p-2} + \ldots x + 1.$$

Theorem 19.1 *$\phi(x)$ is an irreducible polynomial.*

Proof As it stands, the Eisenstein criterion for irreducibility doesn't apply. However a change of variable $y = x - 1$ and the binomial expansion changes the form into

$$\frac{(x^p - 1)}{x - 1} = \frac{((y+1)^p - 1)}{y} = y^{p-1} + py^{p-2} + \frac{p(p-1)}{1 \cdot 2}y^{p-3} + \ldots + p.$$

The binomial coefficients which appear on the right hand side are all integers divisible by the prime p, for p occurs in each numerator as a factor, and can never be cancelled out by the smaller integers in the denominator. The polynomial in y therefore satisfies the Eisenstein criterion and is therefore irreducible – which means that the original equation for $\phi(x)$ is irreducible. ∎

Theorem 19.2 (Irreducibility of $x^p - a = 0$) *A polynomial $f(x) = x^p - a = 0$ of prime degree p over a field K is either irreducible over K or has a root in K.*

Proof 1 *based on the Rational Root Theorem* Let α be a root of $f(x) = x^p - a$ (assume irreducible or not). Let $K = \mathbb{Q}(\alpha)$. (Assume we're working over $\mathbb{Q}$ or $\mathbb{Z}$ by Gauss's lemma for monic polynomials).

Since p prime, $[K : \mathbb{Q}] = 1$ or p.

If =1, $\alpha \in \mathbb{Q}$, so rational root.

Hence if no rational root, $[K : \mathbb{Q}] = p$, so minimum polynomial has degree p, so f (monic) is the minimum polynomial, hence irreducible.

Proof 2 Adjoin to K a primitive p-th root of unity ξ and the a root u of $x^p - a$. The resulting extension $K(\xi, u)$ contains the p roots $u, \xi u, \xi^2 u, \ldots, \xi^{p-1} u$ of the polynomial $x^p - a$, hence is the root field of this polynomial, which has the factorization

$$x^p - a = (x-u)(x-\xi u)(x-\xi^2 u)\ldots(x-\xi^{p-1}u).$$

Suppose that $x^p - a$ has a proper factor $g(x)$ over K of positive degree $m < p$. This factor $g(x)$ is then a product of m of the linear factors of $x^p - a$ over $K(\xi, u)$, so that the constant term b in $g(x)$ is a product of the roots $\xi^i u$. Therefore $b = \xi^k u^m$, for some integer k, and

$$b^p = (\xi^k u^m)^p = (\xi^p)^k (u^p)^m = (u^p)^m = a^m.$$

From this we can find in K a p-th root of a, for $m < p$ is relatively prime to p and there exists integers s and t with $sm + tp = 1$, so that

$$b^p = a^m = a^{1-tp} = a/a^{tp},$$

and $a = (b^s a^t)^p$. The assumed reducibility of $x^p - a$ over K therefore yields a root $(b^s a^t)$ of $x^p - a$ in K. ∎

Theorem 19.3 (Abel's Version of the Irreducibility of $x^p - F$) *If the equations*

$$\begin{aligned} f_1 x^{p-1} + f_2 x^{p-2} + \ldots + f_p &= 0 \\ x^p - F &= 0, \end{aligned} \tag{8}$$

where p is a prime number, are simultaneously satisfied, either $f_1, f_2, f_3, \ldots, f_p$ all vanish, or else one of the roots of $x^p - F = 0$ can be expressed rationally in terms of $f_1, f_2, \ldots, f_p$ and F.

Proof For, suppose the coefficients $f_1, f_2, \ldots, f_p$ don't vanish. The the two equations will have a greatest common divisor (GCD) of the form

$$g(x) = x^k + g_1 x^{k-1} + g_2 x^{k-2} + \ldots + g_k = 0,$$

the coefficients of which are rational functions of $F, f_1, f_2, \ldots, f_p$. Now If x_1 be any one of the roots common to equations (8), the other roots will be of the form

$$\omega^\alpha x_1, \omega^\beta x_1, \ldots, \text{ where } \omega^p = 1.$$

and hence

$$g_k = x_1^k \omega^{\alpha+\beta} \ldots = \omega^\pi x_1^k.$$

Since p is a prime number we can find two numbers m and n of which one is negative, such that $mp + nk = 1$. Also

$$g_k^n = \omega^{n\pi} x_1^{\pi k} = \omega^{n\pi} x_1^{1-nk},$$

and therefore

$$\omega^{n\pi} x_1 = g_k^n F^m,$$

therefore $\omega^{n\pi} x_1$, which is a root of Eq. (8) is expressed rationally in terms of $F, f_1, f_2, \ldots f_k$.

20 Roots of Unity

Let r denote an nth root of unity. If there is no positive integer $m < n$ such that $r^m = 1$, then we say that r is a primitive root nth root of unity.

Theorem 20.1 *Let $R = \cos(2\pi/n) + i\sin(2\pi/n)$. The number R^k, $(1 \leq k \leq n)$ is a primitive nth roof of unity, if and only if, the integers k and n are relatively prime.*

Proof First assume that k and n have a common factor d. Then $n = sd, k = td$. Then

$$(R^k)^s = (R^{td})^s = (R^{sd})^t = (R^n)^t = 1.$$

Since $s < n$, and $(R^k)^s = 1$, it follows that R^k is not a primitive root of unity.

We next assume that k and n are relatively prime, and show that $(R^k)^n$ cannot have the value 1 if $m < n$. We use De Moivre's theorem to write

$$(R^k)^m = \cos\frac{2km\pi}{n} + i\sin\frac{2km\pi}{n}.$$

If the latter quantity is equal to 1, then $2km\pi/n$ must be a multiple of 2π, say $(2km\pi/n) = 2h\pi$. Therefore km must be divisible by n. But this is impossible since k has no factor in common with n and $m < n$ by assumption. Therefore $(R^k)^m \neq 1$ if $m < n$. Therefore R^k is a primitive root of unity if k and n are relatively prime.

21 Root of Unity Filters

Theorem 21.1 *For any polynomial or convergent series $F(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots$, you can filter out the sum of every n-th coefficient $(a_0 + a_n + a_{2n} + \ldots)$ using the n-th roots of unity:*

$$\sum_{k=0}^{\infty} a_{kn} = \frac{1}{n}\sum_{j=0}^{n-1} F(\omega^j)$$

where $\omega = e^{\frac{2\pi i}{n}}$ is a primitive n-th root of unity.

To see exactly why this cancellation happens, we can analyze the sum using the geometric series formula. Let the sum be $S = \sum_{k=0}^{n-1}(\omega^m)^k$, where $\omega = e^{\frac{2\pi i}{n}}$.

Case 1: When m is a multiple of n
If m is a multiple of n (so $m = c \cdot n$ for some integer c):
Each term becomes $(\omega^n)^c$. Since $\omega^n = 1$, each term equals $1^c = 1$. The sum adds 1 exactly n times. Result: $S = 1 + 1 + \cdots + 1 = n$.

Case 2: When m is not a multiple of n
When m is not a multiple of n, the common ratio $r = \omega^m$ is not equal to 1. We apply the finite geometric series formula, $S = \frac{1-r^n}{1-r}$: [1]

Substitute the ratio: $S = \frac{1-(\omega^m)^n}{1-\omega^m}$ Swap the exponents: $S = \frac{1-(\omega^n)^m}{1-\omega^m}$ Evaluate the numerator: Since $\omega^n = 1$, the numerator becomes $1 - 1^m = 0$. Ensure non-zero denominator: Because m is not a multiple of n, $\omega^m \neq 1$, so the denominator is not zero. Result: $S = \frac{0}{\text{non-zero}} = 0$.

Visualizing the Cancellation Geometrically, when m is not a multiple of n, the terms ω^{km} represent vectors pointing symmetrically in all directions around the unit circle. Because of this perfect rotational symmetry, their vectors balance each other out completely, pulling the net sum directly to the origin $(0, 0)$. [1]

The roots of unity filter (or roots of unity trick) is a mathematical technique used in generating functions and combinatorics to isolate and sum specific polynomial coefficients, such as every n-th term. When applied to a product of polynomials, it allows you to easily extract terms whose exponents are periodic. [1, 2]

How It Works
The core of the filter relies on the following summation property for an n-th root of unity ? (where $\omega^n = 1$): [1] $\sum_{k=0}^{n-1} \omega^{mk} = \begin{cases} n, & \text{if } m \text{ is a multiple of } n \\ 0, & \text{otherwise} \end{cases}$

If you have a generating function P(x), you can isolate the sum of every n-th coefficient (e.g., $a_0, a_n, a_{2n}, \ldots$) by averaging P(x) evaluated at all n-th roots of unity: [1, 2] $\sum_{k=0}^{\infty} a_{kn} = \frac{P(1)+P(\omega)+P(\omega^2)+\cdots+P(\omega^{n-1})}{n}$

Applying It to Products In combinatorial problems (such as finding the number of subsets whose sum is divisible by a certain integer), your polynomial is often expressed as a product of factors, $P(x) = \prod_{i=1}^{m} F_i(x)$. [1, 2] Applying the filter to this product involves substituting each root of unity ω^j into the product individually:

Construct the Generating Function: Represent the event or choice as a polynomial, where powers indicate the "value" and coefficients indicate the number of combinations. Evaluate at Roots of Unity: Calculate $P(\omega^j)$ for $j = 0, 1, \ldots, n-1$. Exploit the Product Rule: Because P(x) is a product, $P(\omega^j)$ becomes the product of the individual evaluated factors: $P(\omega^j) = F_1(\omega^j) \cdot F_2(\omega^j) \cdot \cdots \cdot F_m(\omega^j)$

Product of a polynomial over all the roots of unity
When evaluating the product of a polynomial $P(x)$ evaluated over all n-th roots of unity, the result is given by the algebraic formula:

$$P(x_1) \cdot P(x_2) \cdots P(x_n) = P(1) \cdot P(\omega) \cdots P(\omega^{n-1}) = P(n, x^n) \quad (\text{or often simply } (-1)^n P(0)^n)$$

This can be heavily simplified to $x^{mn} - a^n$ for specific forms of polynomials, such as $P(x) = x^m - a$. To calculate this for an arbitrary polynomial $P(x) = a_m x^m + a_{m-1}x^{m-1} + \cdots + a_0$, you can utilize resultants or rely on specific properties of roots of unity, which are formally known to be roots of the polynomial $x^n - 1$. The resulting product depends on the degree of the polynomial and whether the roots include primitive components.
Notice that

$$\begin{aligned} y^p - x^p &= \prod_{i=0}^{p-1} (y - \omega^i x) \text{ therefore} \\ \prod_{i=0}^{p-1} f(\omega^i x) &= Res_y(y^p - x^p, f(y)), \end{aligned}$$

Where $Res_y(h, g)$ is Sylvester's Resolvent for two polynomials $h(x)$ and $g(x)$.

22 Descartes Rule of Signs

Theorem 22.1 *(Descartes Rule of Signs): The number of variatons of sign in the subsequent terms of a polynomial, s, minus the number of positive roots, p, is a non-negative even integer.*

$$s - p = 2r, \quad r \geq 0$$

a) $s - p$ is an even integer.

Let $f(x)$ be a polynomial with a positive leading coefficient .

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_{n-1} x + a_n, \quad \text{and} \quad a_0 > 0.$$

Then since the sequence of signs starts and ends with a $+$ we must have an even number of variations $+-$ and $-+$ of sign in the sequence

$$+ \ldots + .$$

to get back eventually to find the final $+$ sign so s is even.

Similarly p is even because the function must start at $y = a_0$ and can only cross the x-axis an even number of times to eventually again be in the $y > 0$ region. If there are multiple roots of the form $(x - \alpha)^k = 0$, then these are counted with their respective multiplicity. If k is even, then the function does not cross the x-axis at the multiple root. If k is odd, then the function crosses the x axis and goes into the $y < 0$ region. In either case, p is even. Hence if $a_0 > 0$, we have that $s - p$ is even.
Similar arguments hold for the case when $a_0 < 0$. In that case both s and p are odd numbers and therefore $s - p$ is still even.
b) $s - p$ is non-negative.
Use induction on the degree of the polynomial.
$n = 1$. $x + a_n = 0$. Zero variations if a_n is positive and zero positive roots so $s - p = 0$ is even. If $a_n < 0$ then there is one variation in sign and one positive root, $s = 1$ and $p = 1$ and $s - p = 0$.

Induct on $n = k$.

$$f(x) = a_0 x^k + a_1 x^{k-1} + a_2 x^{k-2} + \ldots + a_{k-1} x + a_k.$$

Take the derivative with respect to x,

$$f'(x) = k a_0 x^{k-1} + (k-1) a_1 x^{k-2} + (k-2) a_2 x^{k-3} + \ldots + a_{k-1}.$$

Now, since the k terms that multiply the various x powers are all positive then

$$s = \text{sign variation}[a_0, a_1, a_2, \ldots, a_{k-1}, a_k]$$

and

$$s' = \text{sign variation}[a_0, a_1, a_2, \ldots, a_{k-1}]$$

Therefore

$$s \geq s'.$$

Since the roots p' are interspersed between the roots of $f(x)$ by Rolle's theorem, the number of positive roots can be $p-1$ which occur between the roots of $f(x)$ plus possibly one more between $x=0$ and the first root of α_1 of $f(x)$, so:

$$p' \geq p-1.$$

Therefore we have

$$s \geq s' \geq p' \geq p-1$$

or

$$s-p \geq -1.$$

Part a) shows that $s-p$ must be an even number in all cases, so $s-p \geq -1 \implies s-p \geq 0$. Hence

$$s-p = 2r, \quad r \geq 0.$$

23 Determinants and Their Properties

Consider a set of vectors in an n-dimensional Vector Space: $V = \{\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n\}$ in $\mathbb{R}^n$.

Determinant The Determinant $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ has the following properties

i) Invariance property: $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ is to remain unchanged if some $\boldsymbol{a}_i$ is replaced by $\boldsymbol{a}_i + \boldsymbol{a}_j \quad (i \neq j)$;

ii) Homogeneity property: $D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ is to go over into $\lambda D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n)$ if $\boldsymbol{a}_i$ is replaced by $\lambda \boldsymbol{a}_i$;

iii) the Normalization: $D(\boldsymbol{e}_1, \boldsymbol{e}_2, ..., \boldsymbol{e}_n) = 1$.

The determinant is the unique function of vectors $\{\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n\}$ that satisfy the above properties [3].

Expansion of Determinant Let the S_n be the set of all permutation of the numbers $\{1, 2, ..., n\}$. In the case of n-by-n matrix A we obtain the complete development of the determinant of A as follows:

$$D(\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_n) = \det|A| = \sum_{\sigma \in S_n} \epsilon_{\sigma_1 \sigma_2 \ldots \sigma_n} a_{1\sigma_1} a_{2\sigma_2} \ldots a_{n\sigma_n} = \begin{vmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{i1} & a_{i2} & \ldots & a_{in} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{vmatrix} \tag{9}$$

If the rows are permuted by $\rho \in S_n$, we find

$$\epsilon_{\rho_1 \rho_2 \ldots \rho_n} \det|A| = \sum_{\sigma \in S_n} \epsilon_{\sigma_1 \sigma_2 \ldots \sigma_n} a_{\rho_1 \sigma_1} a_{\rho_2 \sigma_2} \ldots a_{\rho_n \sigma_n} \tag{10}$$

Cofactor of an element a_{rs},

$$M_{rs} = \text{cofactor}(a_{rs}) = \sum_{\sigma \in S_n,\ \sigma_r = s} \epsilon_{\sigma_1 \sigma_2 \ldots \sigma_n} a_{1\sigma_1} a_{2\sigma_2} \ldots a_{r-1\sigma_{r-1}} a_{r+1\sigma_{r+1}} \ldots . a_{n\sigma_n} \tag{11}$$

Relation between cofactors and minors obtained by taking the determinant obtained from A by striking out row r and column s from A. We denote such minor determinants by A_{rs} :

$$M_{rs} = (-1)^{r+s} A_{rs}$$

For every r, $1 \leq r \leq n$ we can write $\det|A|$ in terms of a cofactor expansion

$$\det|A| = \sum_{s=1}^{n} a_{rs}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{rs}\, A_{rs} \tag{12}$$

Also, if there is a mismatch between the row indices between the matrix element and the cofactor or minor, i.e.,

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = 0, \text{ for } \quad k \neq r \tag{13}$$

The case $k \neq r$ corresponds to a determinant with two identical rows and hence vanishes because the rows are linearly dependent. Putting both of the above equations together we find

$$\sum_{s=1}^{n} a_{ks}\, M_{rs} = \sum_{s=1}^{n} (-1)^{r+s}\, a_{ks}\, A_{rs} = \delta_{kr} \det|A| \tag{14}$$

Classical Adjoint and Inverse Matrix We define the Classical Adjoint, also known as the Adjugate, of the matrix A as

$$\mathbf{adj}(A) = M^{\dagger},$$

where $M^{\dagger}$ is the transpose of the cofactor matrix, M. Using Eq.(14), the inverse of A is given by

$$A^{-1} = \frac{1}{|A|}\mathbf{adj}.(A) = \frac{1}{|A|}M^{\dagger}.$$

Since $A^{-1}A = I$ we see that the determinant of A^{-1} is $|A|^{-1}$. Therefore the determinant of the Classical Adjoint matrix is

$$|\mathbf{adj}(A)| = |A|^{n}||A|^{-1} = |A|^{n-1},$$

for an n-by-n matrix A.

Left Inverse and Right Inverse Suppose we have a left inverse X and a right inverse Y of a matrix A. This means that

$$XA = AY = I.$$

Therefore we have

$$\begin{aligned} XAY &= X, \quad \text{and} \\ XAY &= Y. \end{aligned}$$

Therefore

$$X = XAY = Y, \text{ or } X = Y.$$

Therefore left and right inverses of a non-singular matrix must be equal.

The inverse of a Symmetric Matrix is Symmetric A symmetric matrix is defined by $A = A^\dagger$. The inverse of A satisfies

$$A^{-1}A = I$$

Therefore, by taking the transpose of both sides we have

$$A^T(A^{-1})^\dagger = A(A^{-1})^\dagger = I.$$

Therefore

$$A^{-1}A(A^{-1})^\dagger = A^{-1}I = A^{-1},$$

or

$$(A^{-1})^\dagger = A^{-1}.$$

Therefore the inverse of a symmetric matrix is also symmetric.

Special Cases – Inverting a Triangular Matrix

Theorem 23.1 (Upper Triangular Matrices) *Let U be an upper triangular matrix. If the inverse U^{-1} exists, then it is upper triangular.*

Proof Recall the $(n-1)\times(n-1)$ cofactor matrix, M_{rs} that results from omitting row r and column s of $U = (u_{ij})$. When it exists,

$$U^{-1} = \frac{1}{|U|}\mathbf{adj}(U),$$

so it is enough to prove that the $n \times n$ matrix $(|C_{rs}|)$ of cofactor determinants whose transpose is $(|C_{rs}|)^\dagger$ is the adjugate, is lower triangular.

In the case $r < s$, every element below the diagonal of the matrix C_{rs} is also below the diagonal of U, so must be equal to 0.

Hence C_{rs} is upper triangula, with the determinant equal to the product of its diagonal elements.

However, $s - r$ of these diagonal elements are $u_{i+1,j}$ for $i = r, \ldots, s-1$. These elements are from below the diagonal of U, and so are equal to 0.

Hence $r < s$ implies that $|C_{rs}| = 0$, so the $n \times n$ matrix $(|C_{rs}|)$ of cofactor determinants is indeed lower triangular. Hence the adjugate $(|C_{rs}|)^\dagger$ is upper triangular. Hence the inverse of and upper triangular matrix is upper triangular. By similar reasoning we can show that the inverse of a lower triangular matrix is lower triangula. ∎

Laplace's General Expansion Theorem: Divide the set of integers $1, 2, ..., n$ into two complementary sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ where $n = p + q$ and

$$\alpha_1 < \alpha_2 < ... < \alpha_p \text{ and } \beta_1 < \beta_2 < ... < \beta_q.$$

These two sets remained fixed throughout the following:

$$\det A = \sum_{\substack{\rho_1,\rho_2,\ldots,\rho_p \\ \rho_1<\rho_2<\ldots<\rho_p}} (-1)^{(\alpha_1+\alpha_2+\ldots+\alpha_p+\rho_1+\rho_2\ldots+\rho_p)} \begin{vmatrix} a_{\alpha_1\rho_1} & a_{\alpha_1\rho_2} & \cdots & a_{\alpha_1\rho_p} \\ a_{\alpha_2\rho_1} & a_{\alpha_2\rho_2} & \cdots & a_{\alpha_2\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_i\rho_1} & a_{\alpha_i\rho_2} & \cdots & a_{\alpha_i\rho_p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\alpha_p\rho_1} & a_{\alpha_p\rho_2} & \cdots & a_{\alpha_p\rho_p} \end{vmatrix} \begin{vmatrix} a_{\beta_1\sigma_1} & a_{\beta_1\sigma_2} & \cdots & a_{\beta_1\sigma_q} \\ a_{\beta_2\sigma_1} & a_{\beta_2\sigma_2} & \cdots & a_{\beta_2\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_i\sigma_1} & a_{\beta_i\sigma_2} & \cdots & a_{\beta_i\sigma_q} \\ \cdots & \cdots & \cdots & \cdots \\ a_{\beta_q\sigma_1} & a_{\beta_q\sigma_2} & \cdots & a_{\beta_q\sigma_q} \end{vmatrix} \tag{15}$$

The sets $\{\alpha_1, \alpha_2, ..., \alpha_p\}$ and $\{\beta_1, \beta_2, ..., \beta_q\}$ were defined to be fixed. The summation is taken over all p-tuples $(\rho_1, \rho_2, ..., \rho_p)$ from among the numbers $1, 2, ..., n$ for which $\rho_1 < \rho_2 < ... < \rho_p$ and where

$$\sigma_1 < \sigma_2 < ... < \sigma_q$$

are the remaining q integers.
‘

24 Vandermonde Determinant

The Vandermonde Determinant for n variables $x = [x_1, x_2, \ldots, x_n]$ is given by

$$V_n(x) = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^n \\ 1 & x_2 & x_2^2 & x_2^3 & \ldots & x_2^n \\ 1 & x_3 & x_3^2 & x_3^3 & \ldots & x_2^n \\ \vdots & & & & & \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^n \end{bmatrix}$$

Except for the first column, subtract x_1 times the previous column to the rest of the columns. Since a multiple of any column added to any other column does not change the determinant, we can write

$$\det V_n(x) = \det \begin{bmatrix} 1 & 0 & 0 & 0 & \ldots & 0 \\ 1 & x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \ldots & x_2^{n-1}(x_2 - x_1) \\ 1 & x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \ldots & x_3^{n-1}(x_3 - x_1) \\ \vdots & & & & & \\ 1 & x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \ldots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Expanding the determinant using the elements in the first row we have

$$\det V_n(x) = \det \begin{bmatrix} x_2 - x_1 & x_2(x_2 - x_1) & x_2^2(x_2 - x_1) & \ldots & x_2^{n-1}(x_2 - x_1) \\ x_3 - x_1 & x_3(x_3 - x_1) & x_3^2(x_3 - x_1) & \ldots & x_3^{n-1}(x_3 - x_1) \\ \vdots & & & & \\ x_n - x_1 & x_n(x_n - x_1) & x_n^2(x_n - x_1) & \ldots & x_n^{n-1}(x_n - x_1) \end{bmatrix}$$

Factoring out $x_2 - x_1$ from the first row, $x_3 - x_1$ from the second row, and so on until the last row we have

$$\det V_n(x_1, x_2, \ldots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)] \det \begin{bmatrix} 1 & x_2 & x_2^2 & \ldots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \ldots & x_3^{n-1} \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \ldots & x_n^{n-1} \end{bmatrix} = \prod_{i=2}^{n}(x_i - x_1) \ \det V_{n-1}(x_2, x_3, \ldots, x_n)$$

Proceeding in a similar way by subtracting x_2 times the previous column except for the first column we can continue

$$\det V_n(x_1, x_2, \ldots, x_n) = [\prod_{i=2}^{n}(x_i - x_1)] \, [\prod_{i=3}^{n}(x_i - x_2)] \ \det V_{n-2}(x_3, x_4, \ldots, x_n)$$

To arrive at

$$\det V_n(x_1, x_2, \ldots, x_n) = \prod_{\substack{1<j \\ i<j}}^{n}(x_j - x_i) = \prod_{1 \le i < j \le n}(x_j - x_i)$$

25 Resultant of Two Polynomials

Following the demonstration in [7], consider two polynomials

$$f(x) = \sum_{i=0}^{n} f_i\, x^{n-i} = f_0 x^n + f_1 x^{n-1} + f_2 x^{n-2} + \ldots + f_{n-1} x + f_n$$
$$g(x) = \sum_{j=0}^{m} g_j\, x^{m-j} = g_0 x^m + g_1 x^{m-1} + g_2 x^{m-2} + \ldots + g_{m-1} x + g_m$$

Let use assume that $f(x)$ and $g(x)$ have a common factor $(x-\alpha)$. Then

$$f(x) = (x-\alpha) f_1(x) \quad \text{and}$$
$$g(x) = (x-\alpha) g_1(x).$$

Then

$$f(x) g_1(x) = g(x) f_1(x) = \frac{f(x)g(x)}{x-\alpha}.$$

Assuming that

$$f_1(x) = -\sum_{i=0}^{n-1} z_i\, x^{n-1-i} = -(z_0 x^{n-1} + z_1 x^{n-2} + z_2 x^{n-3} + \ldots + z_{n-2} x + z_{n-1})$$
$$g_1(x) = \sum_{j=0}^{m-1} y_j\, x^{m-1-i} = y_0 x^{m-1} + y_1 x^{m-2} + y_2 x^{m-3} + \ldots + y_{m-2} x + y_{m-1}$$

By equating like powers of x, we can write out a linear system corresponding to the equation

$$f(x) g_1(x) - g(x) f_1(x) = 0$$

we find

$$\begin{aligned}
-g(x) f_1(x) = \;& g_0 z_0 x^{n+m-1} + g_0 z_1 x^{n+m-2} + g_0 z_2 x^{n+m-3} + \ldots + g_0 z_{n-2} x^{m+1} + g_0 z_{n-1} x^m + \\
& g_1 z_0 x^{n+m-2} + g_1 z_1 x^{n+m-3} + \ldots + g_1 z_{n-3} x^{m+1} + g_1 z_{n-2} x^m + g_1 z_{n-1} x^{m-1} + \\
& g_2 z_0 x^{n+m-3} + g_2 z_1 x^{n+m-4} + \ldots + g_2 z_{n-3} x^m + g_2 z_{n-2} x^{m-1} + g_2 z_{n-1} x^{m-2} + \\
& \vdots \;\; + \\
& \ldots + (g_{m-2} z_{n-1} + g_{m-1} z_{n-2} + g_m z_{n-3}) x^2 + (g_{m-1} z_{n-1} + g_m z_{n-2}) x + g_m z_{n-1}
\end{aligned}$$

$$\begin{aligned}
g(x) g_1(x) = \;& f_0 y_0 x^{n+m-1} + f_0 y_1 x^{n+m-2} + f_0 y_2 x^{n+m-3} + \ldots + f_0 y_{m-2} x^{n+1} + f_0 y_{n-1} x^n + \\
& f_1 y_0 x^{n+m-2} + f_1 y_1 x^{n+m-3} + \ldots + f_1 y_{n-3} x^{n+1} + f_1 y_{m-2} x^n + f_1 y_{n-1} x^{n-1} + \\
& f_2 y_0 x^{n+m-3} + f_2 y_1 x^{n+m-4} + \ldots + f_2 y_{n-3} x^n + f_2 y_{m-2} x^{n-1} + f_2 y_{n-1} x^{n-2} + \\
& \vdots \;\; + \\
& \ldots + (f_{n-2} y_{m-1} + f_{n-1} y_{m-2} + f_n y_{m-3}) x^2 + (f_{n-1} y_{m-1} + f_n y_{m-2}) x + f_n y_{m-1}
\end{aligned}$$

In order to satisfy $f(x)g_1(x) - g(x)f_1(x) = 0$, all the terms multiplying like powers of x each must vanish. This can be arranged as a linear system in the vector $[z_0, z_1, ..., z_{n-1}, y_0, y_2, ...y_{m-1}]$. The coefficient matrix of this linear system is equal to the transpose of the Sylvester Resultant matrix which is defined as

$$R_{m,n}(g,f) = \underbrace{\begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix}}_{m+n} \begin{matrix} \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} n \text{ lines} \\ \left.\vphantom{\begin{matrix} a \\ a \\ a \end{matrix}}\right\} m \text{ lines} \end{matrix}$$

The corresponding linear system that sets all the coefficients of the same powers of x to 0 in $f(x)g_1(x)-g(x)f_1(x) = 0$ is

$$\begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix}^T \begin{bmatrix} z_0 \\ \vdots \\ z_{n-1} \\ y_0 \\ \vdots \\ y_{m-1} \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The criterion for a non-trivial solution to the above linear system is

$$\det \begin{bmatrix} g_0 & \cdots & g_m & & \\ & \ddots & \cdots & \ddots & \\ & & g_0 & \cdots & g_m \\ f_0 & \cdots & f_n & & \\ & \ddots & \cdots & \ddots & \\ & & f_0 & \cdots & f_n \end{bmatrix} = 0$$

The rank of the matrix $R_{m,n}(g, f)$ also gives the multiplicity of the common root of $f(x)$ and $g(x)$. The leading term of this determinant is $g_0^n f_n^m$. The terms of in the expansion an $(n + m) \times (n + m)$ matrix M of the form $\sum_\sigma \epsilon_{r_1,r_2,r_3,...r_{n+m}} m_{1r_1} m_{2r_2} m_{3r_3} \ldots m_{(n+m)r_{n+m}}$ where the sum is over all permutations σ of the numbers $[1, 2, \ldots, n+m]$ and $sgn(\sigma) = \epsilon_{r_1,r_2,r_3,...r_{n+m}}$ is the sign of the permutation. In the case of the Sylvester Resultant the terms take the form

$$sgn(\sigma) g_0^{\nu_0} g_1^{\nu_1} \ldots g_m^{\nu_m} f_0^{\mu_0} f_1^{\mu_1} \ldots f_n^{\mu_n},$$

where

$$n = \nu_0 + \nu_1 + \nu_2 + \ldots + \nu_m$$

$$m = \mu_0 + \mu_1 + \mu_2 + \ldots + \mu_n$$

Factoring out $g_0^n f_0^m$ the generic term in the expansion of the determinant becomes

$$sgn(\sigma) g_0^n f_0^m (\frac{g_1}{g_0})^{\nu_1} (\frac{g_2}{g_0})^{\nu_2} \ldots (\frac{g_m}{g_0})^{\nu_m} (\frac{f_1}{f_0})^{\mu_1} (\frac{f_2}{f_0})^{\mu_2} \ldots (\frac{f_n}{f_0})^{\mu_n}$$

Hence the Sylvester Resultant has the form

$$R_{m,n}(g,f) = g_0^n f_0^m \phi(\alpha,\beta)$$

where ϕ is an integral function of the roots α_i and β_i. As the determinant vanishes for any common roots $\alpha_i = \beta_k$, the determinant is divisible by $(\alpha_i - \beta_k)$ and the determinant is divisible by

$$g_0^n f_0^m \prod_{i,k}(\alpha_i - \beta_k)$$

Theorem 25.1 *$R_{m,n}(g,f)$ is a homogenous polynomial with integer coefficients g_i, f_j.*

(i) $R_{m,n}(g,f)$ is homogenous of degree n in $g_m, ..., g_0$ and degree m in $f_n, ..., f_0$.

(ii)If g_i and f_i are regarded as having degree i. then $R_{m,n}(g,f)$ is homogenous of degree mn.

Proof It is obvious that $R_{m,n}(g,f)$ is a homogenous polynomial with integer coefficients in the coefficients g_i, f_j, of total degree $m+n$. Moreover, to replace g_i by tg_i and f_j by uf_j in the Sylvester matrix means that we multiply each of the first m rows by t and each of the last n rows by u, and thus the determinant $R_{m,n}(g,f)$ by $t^m u^n$, which shows part (i). It is here best to treat g_i, f_j, t and u as different inderminants a do the calculations in $F(g_0, ..., g_m, f_0, ..., f_n, t, u)$. Similarly, (ii) follows because to replace each g_i by $t^i g_i$ and each f_j by $t^j f_j$ in $R_{m,n}(g,f)$ yields the same result as multiplying the ith row by t^{m+i} for $i = 1, 2, ..., m$ and by t^i for $i = m+1, ..., m+n$, and the jth column by t^{-j}. The overall effect it to multiply the determiant $R_{m,n}(g,f)$ by t^K where $K = mn + \sum_{i=1}^{m+n} i - \sum_{j=1}^{m+n} j = mn$. ∎

26 Application to Clearing Denominators or Irrationalities

Sylvester's dialytic elimination method (and the resulting dialytic resolvent) can be used as an alternate approach to clear denominators and eliminate variables, yielding the same final polynomial equations as Abel's method of multiplying conjugate algebraic expressions to clear denominators of radical expressions and roots of unity. While Niels Henrik Abel's famous 1826 paper on the impossibility of solving the general quintic equation relied on multiplying by algebraic conjugates to clear denominators systematically, James Joseph Sylvester's later dialytic method (developed in the 1840s) reformulates the elimination problem into linear algebra. Instead of constructing field extensions and tracking field conjugates, Sylvester's approach creates a system of linear equations by treating powers of the variable as independent unknowns, solving it via a single determinant
1. Formulated the Algebraic Elimination Suppose Abel's method encounters an expression involving an algebraic denominator, such as $y = \frac{A(x)}{B(x)}$, where x is a root of a known polynomial $P(x) = 0$. Abel would clear the denominator by finding the algebraic conjugates of $B(x)$ relative to the roots of $P(x)$ and multiplying them.

To use Sylvester's resolvent instead, we rewrite the relationship as a system of two polynomial equations sharing a common root x:$P(x) = 0$ (The base polynomial equation)$R(x,y) = y \cdot B(x) - A(x) = 0$ (The relation defining y) Our goal is to eliminate x entirely to find a polynomial equation purely in terms of y.

2. Construct the Dialytic Matrix Sylvester's method eliminates x by generating a set of linearly independent equations. If $P(x)$ has degree n and $R(x,y)$ has degree m with respect to x, we multiply $P(x)$ by successive powers of x up to x^{m-1}, and multiply $R(x,y)$ by powers of x up to x^{n-1}. This creates a system of $n+m$ equations:$x^k \cdot P(x) = 0 \quad \text{for } k = 0, 1, \ldots, m-1 x^j \cdot R(x,y) = 0 \quad \text{for } j = 0, 1, \ldots, n-1$ We treat each power of x

(from x^0 up to x^{n+m-1}) as a distinct, independent linear variable.

3. Compute the Sylvester Resolvent For a non-trivial solution to exist for this linear system, the determinant of the coefficients must equal zero. This determinant is the Sylvester Resultant (or Sylvester Resolvent), denoted as $\mathrm{Res}(P, R, x)$. The matrix takes the form:

$$\mathrm{Res}(P,R,x) = \det \begin{pmatrix} p_n & p_{n-1} & \cdots & p_0 & 0 & \cdots & 0 \\ 0 & p_n & p_{n-1} & \cdots & p_0 & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \\ r_m(y) & r_{m-1}(y) & \cdots & r_0(y) & 0 & \cdots & 0 \\ 0 & r_m(y) & r_{m-1}(y) & \cdots & r_0(y) & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \end{pmatrix} = 0$$

Evaluating this determinant yields a pure polynomial equation $Q(y) = 0$, completely clearing the denominator without ever needing to calculate explicit field conjugates or Galois orbits.

Conclusion Sylvester's dialytic resolvent acts as a perfect mathematical alternative to Abel's conjugate multiplication for clearing denominators. By framing the algebraic relation as a simultaneous polynomial system, Sylvester bypasses the need to explicitly compute field conjugates, instead achieving the exact same polynomial reduction through the evaluation of a single matrix determinant.
To see exactly how Sylvester's method serves as a direct alternative, we can look at a representative reduction from Niels Henrik Abel's work. In his 1826 impossibility proof, Abel builds algebraic functions of increasing"orders" by adjoining radicals. He routinely deals with expressions where a new variable y is a rational function of a root x, where x satisfies a lower-degree base polynomial equation. Let's isolate a core algebraic step typical of Abel's paper: reducing an expression where y is defined by a denominator containing a quadratic root x, where x satisfies the base cubic equation:$P(x) = x^3 + ax + b = 0$ Suppose the relationship defining the new variable y is given by the rational function:$y = \frac{1}{x^2+x}$

1. Formulate the Two Polynomials To clear the denominator using Sylvester's method instead of multiplying by algebraic conjugates, we first transform the rational function into a second polynomial equation, $R(x, y) = 0$, grouped by powers of x:$y(x^2 + x) - 1 = 0 \implies R(x, y) = yx^2 + yx - 1 = 0$Now we have a system of two polynomial equations sharing the common root x:$P(x) = 1x^3 + 0x^2 + ax + b = 0$ (Degree $n = 3$)$R(x, y) = yx^2 + yx - 1 = 0$ (Degree $m = 2$ with respect to x)

2. Set Up the Rows of the Sylvester Matrix The Sylvester matrix will have a total dimension of $(n+m)\times(n+m)$, which means a 5×5 matrix. We shift the coefficients of each polynomial across the columns:Rows 1 & 2 (from $P(x)$): Multiply $P(x)$ by x^1 and x^0. Rows 3, 4, & 5 (from $R(x, y)$): Multiply $R(x, y)$ by x^2, x^1, and x^0. The columns correspond to the descending independent powers of x: $\begin{pmatrix} x^4 & x^3 & x^2 & x^1 & x^0 \end{pmatrix}$.

3. The Explicit Sylvester Matrix $\mathbf{S} = \begin{pmatrix} 1 & 0 & a & b & 0 \\ 0 & 1 & 0 & a & b \\ y & y & -1 & 0 & 0 \\ 0 & y & y & -1 & 0 \\ 0 & 0 & y & y & -1 \end{pmatrix}$

4. Clearing the Denominator To completely eliminate x and clear the denominator, you take the determinant of this matrix and set it to zero ($\det(\mathbf{S}) = 0$).Evaluating this 5×5 determinant yields a pure polynomial equation in terms of y, a, and b. This polynomial gives you the exact same result Abel would get by finding the other conjugate roots of the cubic equation and computing: $(y(x_1^2 + x_1) - 1)(y(x_2^2 + x_2) - 1)(y(x_3^2 + x_3) - 1) = 0$

27 Abel's Method of Rationalizing a Ratio

As an application of Sylvester's Resultant, let us apply the theorem to the case of two polynomials which are related to Abel's Proof of the insolubility of the quintic equation in terms of radicals. In particular let us examine Abel's method of rationalizing a polynomial function of radicals. Let

$$\begin{aligned} f(x) &= x^p - R \\ g(x) &= a_{p-1}x^{p-1} + a_{p-2}x^{p-2} + \ldots + a_2x^2 + a_1x + a_0, \end{aligned}$$

where R is not a pth power of an element in the field of coefficients. $f(x)$ and $g(x)$ will have a common root if the Sylvester Resultant formed by these two functions vanishes. The roots of $f(x) = 0$ take the form $z_k = \omega^k R^{\frac{1}{p}}$ and we're assuming that $R^{\frac{1}{p}}$ is an irrationality in the domain containing the coefficients. Also with respect to Abel's rationalization method we need to show that

$$\prod_{k=0}^{p-1} g(z_k) = G(z^p) = G(R) \text{ where } G(R) \text{ is rational in the domain of the coefficients of } g(x). \tag{16}$$

Using $f(x)$ and $g(x)$ we construct the Sylvester Resultant

$$\det(M) = \prod_{k=0}^{p-1} g(z_k) = \det \begin{bmatrix} 1 & 0 & \cdots & -R & & \\ 0 & 1 & \ddots & \cdots & \ddots & \\ \vdots & & \ddots & \cdots & \ddots & \\ 0 & \cdots & & 1 & \cdots & -R \\ a_{p-1} & & \cdots & a_0 & & \\ & & \ddots & \cdots & \ddots & \\ & & & a_{p-1} & \cdots & a_0 \end{bmatrix}.$$

Since the roots of $f(x) = x^p - R$ are exactly $z_k = \omega^k R^{\frac{1}{p}}$, substituting them into the formula yields

$$\det(M) = \prod_{k=0}^{p-1} g(z_k).$$

Because computing the determinant $\det(M)$ only involves additions, subtractions, and multiplication of the matrix entries, the final polynomial expression outputted by the determinant can only contain powers of combinations of the terms in the matrix M. Therefore the resulting polynomial must be a function of $x^p = R$ and the coefficients and must therefore be a rational expression. The above is a demonstration of Abel's method of Rationalization.

28 Substitutions and Almost Symmetric Functions

Substition refers to the set of permutations of n elements which form the Symmetric Group.

Theorem 28.1 *All the substitutions of $x_1, x_2, \ldots, x_n$ which leave unaltered a rational function $\phi(x_1, x_2, \ldots, x_n$ form a group G.*

Let ϕ_s denote the function obtained by applying to ϕ the substitusion s. If a and b are two substitutions which leave ϕ unaltered, then $\phi_a = \phi, \phi_b = \phi$. Then

$$(\phi_a)_b = (\phi)_b = \phi_b = \phi, \text{ or } \phi_{ab} = \phi$$

Hence the product of ab is one of the substitutions which leave ϕ unaltered and the set G has the group property.

The group G is called the *group of the function ϕ*, while ϕ is said to *belong to the group G*

Theorem 28.2 *Every substitution can be expressed as a product of transpositions in various ways.*

Theorem 28.3 *Of the various decompositions of a given substitution s into a product of transpositions, all contain an even number of transpositions (an even substitution), or all constain an odd number of transpositions (an odd substitution).*

Corollary 28.4 *The totality of even substitutions on n letters forms a group, called the Alternating Group on n letters.*

Consider n elements $x = \{x_1, x_2, x_3, \ldots, x_n\}$. Consider the rational function

$$\begin{array}{rlll}\phi(x) = \prod_{i<j}(x_i - x_j) = & (x_1 - x_2)(x_1 - x_3)(x_1 - x_4) & \ldots\ldots & (x_1 - x_n) \\ & (x_2 - x_3)(x_2 - x_4) & \ldots\ldots & (x_2 - x_n) \\ & (x_3 - x_4) & \ldots\ldots & (x_3 - x_n) \\ & & \ldots\ldots\ldots & \\ & & & (x_{n-1} - x_n)\end{array}$$

consisting of the product of all the difference of the elements.

$\phi(x)$ is related to the Vandemonde determinant

$$\phi(x) = V(x) = \begin{vmatrix} 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & x_2^3 & \ldots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & x_3^3 & \ldots & x_2^{n-1} \\ \vdots & & & & & \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^{n-1} \end{vmatrix}$$

The square of $V(x)$ is a well- known symmetric function, the discriminant $\triangle$. Therefore $V(x)$ has two values equal numerically with opposites signs, viz. $\sqrt{\triangle}$ and $-\sqrt{\triangle}$. $V(x)$ is an example of an alternating function. Any transposition alters the sign of $V(x)$. For example, the transposition $(x_1 x_2)$ affects only the terms in the first and second lines of the product, and replaces them by

$$\begin{array}{rll}(x_2 - x_1)(x_2 - x_3)(x_2 - x_4) & \ldots\ldots & (x_2 - x_n) \\ (x_1 - x_3)(x_1 - x_4) & \ldots\ldots & (x_1 - x_n).\end{array}$$

Hence, if s is the product of an even number of transpositions, it leaves $v(x)$ unaltered; if s is the product of an odd number of transpositions, it changes $V(x)$ into $-V(x)$.

29 Lagrange

Theorem 29.1 (Lagrange) *The order of a subgroup is a divisor of the order of the group.*

Proof Consider a group G of N and a subgroup H composed of the substitutions

$$h_1 = I, h2, h3, \ldots, h_P. \tag{17}$$

If G constains no further substitutions, then $N = P$ and the theorem is true. Now let G contain a substitution g_2 not in H. Then G also must contain

$$g_2, h_2g_2, h_3g_2, \ldots, h_Pg_2. \tag{18}$$

This set are all distinct and all different from the substitutions (17), since $h_\alpha g_2 = h_\beta$ requires that $g_2 = h_\alpha^{-1}h_\beta =$ a substition of H contrary to the assumption that g_2 is distinct from the elements of H. Hence the substitutions combined now give $2P$ distinct substitutions of G. If there are no other substitutions in G, then $N = 2P$ and the theorem is true. If there are any other distrinct substitutions g_3 in G not in (17) or (18), then G contains

$$g_3, h_2g_3, h_3g_3, \ldots, h_Pg_3. \tag{19}$$

Moreover, these substitutions are all different from (17) and (18), since $h_\alpha g_3 = h_\beta g_2$ requires that $g_3 = h_\alpha^{-1}h_\beta g_2$ and shall belong to (18), contrary to assumption. We now have $3P$ distinct substitutions of G. Either $N = 3P$ or else G contains a substitution g_4 not in the sets (17), (19), (19). In the latter case G contains the products

$$g_4, h_2g_4, h_2g_4, \ldots, h_Pg_4, \tag{20}$$

all of which are distrinct and all different from the sets (17), (19), (19), so that we have $4P$ distinct substitutions. Proceeding in this way, we finally reach a last set of P substitutions

$$g_nu, h_2g_nu, h_3g_nu, \ldots, h_Pg_\nu, \tag{21}$$

since the order of H is finite. Hence $N = \nu P$.

Definition The number ν is called the *index* of the subgroup H under G and is denoted by $\nu = [G : H]$.

Corollary 29.2 *The order of any group H of substitutions on n letters is a divisor of $n!$. H is a subgroup of the symmetric group G_n on n letters.*

Theorem 29.3 *The period of any substitution contained in a group G of order N is a divisor of N.*

Corollary 29.4 (Applied to the Symmetric Group) *The order of any substitution group on a set of n elements is an exact divisor of $N = n!$, the quotient being the number of distinct value of the corresponding multiple-valued function.*

Let us arrange the N substitutions of G in a rectangular array using the substitutions of a subgroup H in the first row:

$$\begin{array}{lllll} h_1 = I & h_2 & h_3 & \dots & h_P \\ g_2 & h_2g_2 & h_3g_2 & \dots & h_Pg_2 \\ g_3 & h_2g_3 & h_3g_3 & \dots & h_Pg_3 \\ \vdots & \dots & \dots & \dots & \vdots \\ g_\nu & h_2g_\nu & h_3g_\nu & \dots & h_Pg_\nu \end{array} \tag{22}$$

The $g_1 = I, g_2, g_3, \dots, g_\nu$ multiply the elements of H on the right hand side forming right cosets. Similarly, a rectangular array for the substitiions could be formed by using left costs.

Theorem 29.5 *If ψ is a rational funciton of $x_1, x_2, \dots, x_n$ belonging to a subgroup H of index ν, then ψ is ν-valued under the full set of substitutions of G.*

Proof Apply to ψ all the N substitutions of G arranged in a rectangular array as in (22). All the substitutions belonging to the same row give the same answer since

$$\psi_{h_i g_\alpha} = (\psi_{h_i})_{g_\alpha} = (\psi)_{g_\alpha} = \psi_{g_\alpha}.$$

Hence there result at most ν values. But, if

$$\psi_{g_\alpha} = \psi_{g_\beta}$$

then $\psi_{g_\alpha g_b eta^{-1}} = \psi$, so that $g_\alpha g_\beta^{-1} h_i \in H$, leaving ψ unaltered. Hence $g_\alpha = h_i g_\alpha$, contrary to the assumption made in forming the rectangular array.

Regarding the orders of a group H and a larger group G containing H as a subgroup; that is to say, the order r of H is an exact divisor of the order G which contains H as a subgroup, the quotient m being the number of distinct values of a multiple-valued function unaltered by the substitutions of G which are obtained by the substutions of H.

Definition The ν distinct functions $\psi, \psi_{g_2}, \psi_{g_3}, \dots, \psi_{g_\nu}$ are called **conjugate values** of ψ under G.

Theorem 29.6 *The ρ distinct values which a rational function $\psi(x_1, \dots, x_n)$ takes when operated on by all $n!$ substitutions are the roots of an equation of degree ρ whose coefficients are rational functions of the elementary symmetric functions.*

Formation of Functions of a Given Group

We define the Galois Function having $n!$ values for all the substiutionts of the symmetric group

$$\psi_1 = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \dots \alpha_n x_n,$$

in which $\alpha_1, \alpha_2, \dots, \alpha_n$ are n distinct arbitrary constants. This function is called the Galois Function. If we use all the P substitutions of H to form the functions $\psi_1, \psi_2, \dots, \psi_P$ by the substitutions of H anysymmetric funciton of these values will be unaltered by the substitutions of H. In particular $\phi_1 = (y - \psi_1)(y - \psi_2) \dots (y - \psi_P)$ will be unaltered by the substutions of H, and altered to a different value by the substitutions not in H. The function ϕ_1 is not therefore unaltered by the substitutions of any wider group that contains H as a subgroup. Noting the forms of the expression of the sums of powers of the roots in terms of the coefficients, we see that instead of the coefficients of the powers of y in ϕ_1 we may take sums of powers of $\psi_1, \psi_2, \dots, \psi_P$ up to the P^{th} and deduce that one at least of such P sums of powers will provide a function unlatered by the substitutions of H and altered by any other substitution of the symmetric group.

Theorem 29.7 *Every integral symmetric function of the distinct values of any integral multiple-valued function of n elements is a symmetric function of the elements themselves.*

Proof Let $F(\phi_1, \phi_2, \ldots, \phi_\rho)$ be any rational integral symmetric function of the ρ-values. Any substitution S whatever affecting the elements applied to these ρ-values either leaves any function unchanged or replaces it by one of the others. No two of the resulting values can be equal, for if $\phi_i S$ were equal to $\phi_j S$ it would follow, by applying the substitution S^{-1}, that $\phi_i = \phi_j$ which is contrary to the assumptions. Consequently the ρ values of ϕ are reproduced by S in some order or other. The symmetric function F therefore remains unchanged by any substitutiont, and is consequently a symmetric element of the elements themselves.

Corollary 29.8 *The ρ distinct values of an integral multiple-valued function are roots of an equation whose coefficients are integral symmetric functions of the elements themselves.*

Theorem 29.9 (Lagrange Theorem on functions belonging to the same group) *Two rational functions belonging to the same group are rationally expressed each in terms of the other.*

Proof Let ϕ_1 and ψ_1 be two functions belonging to the same group

$$H = \{I, S_2, S_3, \ldots, S_r\},$$

of order r and degree n, each of these functions having ρ distinct values, where $r * \rho = N$. Any substitution not contained in H will convert ϕ_1 into another of its values, say ϕ_k and at the same time convert ψ_1 into ψ_k. By operating all possible substitutions, ρ pairs of values $\phi_1, \psi_1 : \phi_2, \psi_2 : \ldots \phi_\rho, \psi_\rho$ are obtained. Now, in the first place, the rational function

$$\sum \phi^i \psi^j = \phi_1^i \psi_1^j + \phi_2^i \psi_2^j + \ldots + \phi_k^i \psi_k^j + \ldots + \phi_\rho^i \psi_\rho^j \tag{23}$$

is clearly a symmetric function of the elements, for it appears by the same reasoning as before that any substitution acting on this function will merely permute the terms leaving the sum unaltered. Therefore $\sum \phi^i \psi^j$ is a symmetric function of the elements.

If we now take $j = 1$ and assign to i all the values $0, 1, 2, \ldots, \rho - 1$ in succession, we obtain the following ρ equations linear in $\psi_1, \psi_2, \ldots, \psi_\rho$:

$$\begin{array}{rcrcccrl}
\psi_1 & + & \psi_2 & + & \ldots & + & \psi_\rho = & T_0 \\
\phi_1\psi_1 & + & \phi_2\psi_2 & + & \ldots & + & \phi_\rho\psi_\rho = & T_1 \\
\phi_1^2\psi_1 & + & \phi_2^2\psi_2 & + & \ldots & + & \phi_\rho^2\psi_\rho = & T_2 \\
\ldots & \ldots & \ldots & \ldots & \ldots & \ldots & \ldots = & \ldots \\
\phi_1^{\rho-1}\psi_1 & + & \phi_2^{\rho-1}\psi_2 & + & \ldots & + & \phi_\rho^{\rho-1}\psi_\rho = & T_{\rho-1}
\end{array} \tag{24}$$

where $T_0, T_1, \ldots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \ldots, x_n$.

We can write the above as a linear system

$$\begin{bmatrix}
1 & 1 & 1 & 1 & \ldots & 1 \\
\phi_1 & \phi_2 & \phi_3 & \phi_4 & \ldots & \phi_\rho \\
\phi_1^2 & \phi_2^2 & \phi_3^2 & \phi_4^2 & \ldots & \phi_\rho^2 \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
\phi_1^{\rho-1} & \phi_2^{\rho-1} & \phi_3^{\rho-1} & \phi_4^{\rho-1} & \ldots & \phi_\rho^{\rho-1}
\end{bmatrix}
\begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \vdots \\ \psi_\rho \end{bmatrix}
=
\begin{bmatrix} T_0 \\ T_1 \\ \vdots \\ \vdots \\ T_{\rho-1} \end{bmatrix} \tag{25}$$

Linear systems such as Eq.(25) can be solved using Cramer's rule. Alternatively, in order to understand better the relatonship between ϕ_i and ψ_i let us consider a method based on transforming non-homogenous linear systems into sets of corresponding homogeneous linear systems in each of the unknown parameters. To this effect consider the linear system [4].

$$\begin{array}{ccccccc} x & + & y & + & z & = & T_0 \\ \alpha x & + & \beta y & + & \gamma z & = & T_1 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & T_2 \end{array} \tag{26}$$

By adding a fourth equation to the above we can isolate one of the variables, say y, and determine the functional dependence of y on β and the symmetric functions of the coefficients. Adding $y = y$ to the above system and converting it into a system of homogenous equations by introducing a 4th unknown w we obtain the system

$$\begin{array}{ccccccc} 0 & + & y & + & 0 & = & y \\ x & + & y & + & z & = & T_0 \\ \alpha x & + & \beta y & + & \gamma z & = & T_1 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & = & T_2 \end{array} \tag{27}$$

The above system can be converted into a Homogeneous System by moving the final column to the left-hand side of the equal sign.

$$\begin{array}{ccccccccc} 0 & + & y & + & 0 & - & y & = & 0 \\ x & + & y & + & z & - & T_0 & = & 0 \\ \alpha x & + & \beta y & + & \gamma z & - & T_1 & = & 0 \\ \alpha^2 x & + & \beta^2 y & + & \gamma^2 z & - & T_2 & = & 0 \end{array} \tag{28}$$

Writing the above system using a matrix of coefficients and introducing the 4th undetermined variable we find

$$\begin{bmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -T_0 \\ \alpha & \beta & \gamma & -T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -T_2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{29}$$

In order for this homogenous system to be consistent with Eq.(26) the determinant of the matrix of coefficients must vanish.

$$\begin{vmatrix} 0 & 1 & 0 & -y \\ 1 & 1 & 1 & -T_0 \\ \alpha & \beta & \gamma & -T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & -T_2 \end{vmatrix} \tag{30}$$

Since the determinant vanishes we can factor out the -1 from the last column and write the consistency condition as

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & T_0 \\ \alpha & \beta & \gamma & T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & T_2 \end{vmatrix} = 0. \tag{31}$$

Multiplying the above determinant by the non-singular determinant

$$\begin{vmatrix} 1 & 1 & 1 & 0 \\ \alpha & \beta & \gamma & 0 \\ \alpha^2 & \beta^2 & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \tag{32}$$

we find

$$\begin{vmatrix} 0 & 1 & 0 & y \\ 1 & 1 & 1 & T_0 \\ \alpha & \beta & \gamma & T_1 \\ \alpha^2 & \beta^2 & \gamma^2 & T_2 \end{vmatrix} \begin{vmatrix} 1 & \alpha & \alpha^2 & 0 \\ 1 & \beta & \beta^2 & 0 \\ 1 & \gamma & \gamma^2 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & \beta & \beta^2 & y \\ s_0 & s_1 & s_2 & T_0 \\ s_1 & s_2 & s_3 & T_1 \\ s_2 & s_3 & s_4 & T_2 \end{vmatrix} = 0, \tag{33}$$

where

$$s_i = \alpha^i + \beta^i + \gamma^i$$

and the s_i are symmetric functions in $\alpha, \beta,$ and γ.

Since the s_i's are sums of powers of α, β, γ and are symmetric functions, they can be expressed rationally in terms of the elementary symmetric functions. By expanding the final determinant in Eq. (33) along the top row and setting the expansion equal to zero to satisfy the consistency requirement we see that we have y expressed as a quadratic function of β along with the sums of powers of the three quantities α, β, γ. Summarizing the results obtained – we expand the above determinant along the top row and solve for y.

$$y = \frac{\begin{vmatrix} s_1 & s_2 & T_0 \\ s_2 & s_3 & T_1 \\ s_3 & s_4 & T_2 \end{vmatrix} - \beta \begin{vmatrix} s_0 & s_2 & T_0 \\ s_1 & s_3 & T_1 \\ s_2 & s_4 & T_2 \end{vmatrix} + \beta^2 \begin{vmatrix} s_0 & s_1 & T_0 \\ s_1 & s_2 & T_1 \\ s_2 & s_3 & T_2 \end{vmatrix}}{\begin{vmatrix} s_0 & s_1 & s_2 \\ s_1 & s_2 & s_3 \\ s_2 & s_3 & s_4 \end{vmatrix}}. \tag{34}$$

The right-hand-side depends quadratically on β and other expressions which are symmetric functions of $\alpha, \beta,$ and γ and hence demonstrate the claim made in Theorem (29.9).

The denominator of Eq. (34) is a function of the squares of the differences of $\alpha, \beta,$ and γ:

$$\triangle(\alpha, \beta, \gamma) = \begin{vmatrix} s_0 & s_1 & s_2 \\ s_1 & s_2 & s_3 \\ s_2 & s_3 & s_4 \end{vmatrix} = (\alpha - \beta)^2 (\alpha - \gamma)^2 (\beta - \gamma)^2 \tag{35}$$

Using this result we can write the solution for y as

$$\triangle(\alpha, \beta, \gamma) y = \begin{vmatrix} s_1 & s_2 & T_0 \\ s_2 & s_3 & T_1 \\ s_3 & s_4 & T_2 \end{vmatrix} - \beta \begin{vmatrix} s_0 & s_2 & T_0 \\ s_1 & s_3 & T_1 \\ s_2 & s_4 & T_2 \end{vmatrix} + \beta^2 \begin{vmatrix} s_0 & s_1 & T_0 \\ s_1 & s_2 & T_1 \\ s_2 & s_3 & T_2 \end{vmatrix}. \tag{36}$$

Returning to the original problem Eq. (24), we can similarly express the solution for ψ_j in terms of ϕ_j in the form of a determinant:

$$\begin{vmatrix} 1 & \phi_j & \phi_j^2 & \dots & \phi_j^{\rho-1} & \psi_j \\ s_0 & s_1 & s_2 & \dots & s_{\rho-1} & T_0 \\ s_1 & s_2 & s_3 & \dots & s_\rho & T_1 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ s_{\rho-1} & s_\rho & s_{\rho+2} & \dots & s_{2\rho-2} & T_{\rho-1} \end{vmatrix} = 0, \tag{37}$$

where $T_0, T_1, \dots, T_{\rho-1}$ are all symmetric functions of $x_1, x_2, \dots, x_n$ and $s_k = \phi_1^k + \phi_2^k + \phi_3^k + \dots + \phi_\rho^k$.

The solution of these equations takes the form

$$\triangle(\phi_1, \phi_2, \phi_3, \ldots, \phi_\rho)\psi_j = A_0\phi_j^{\rho-1} + A_1\phi_j^{\rho-2} + \ldots + A_{\rho-1}, \tag{38}$$

where $\triangle$ is the product of the square of the differences of all the ϕ's (the square of the Vandermonde determinant) and $A_0, A_1, \ldots, A_{\rho-1}$ are all symmetric in $x_1, x_2, \ldots, x_n$.

Therefore ψ_j can be rationally expressed as a rational function of ϕ_j and the elementary symmetric functions of $x_1, x_2, \ldots, x_n$ (the coefficients of the polynomial that has $x_1, x_2, \ldots, x_n$ as roots).

Lagrange Interpolation Formula:
Given a set of $k+1$ nodes $\{x_0, x_1, \ldots, x_k\}$, which must all be distinct, $x_j \neq x_m$ for indices $j \neq m$, the *Lagrange basis* for polynomials of degree $\leq k$ for those nodes is the set of polynomials $\{l_0(x), l_1(x), \ldots, l_k(x)\}$ each of degree k which take values $l_j(x_m) = 0$ if $m \neq j$ and $l_j(x_j) = 1$. Using the Kronecker delta this can be written $l_k(x_m) = \delta_{jm}$. Each basis polynomial can be explicitly described by the product:

$$\begin{aligned} l_j(x) &= \frac{(x-x_0)}{(x_j-x_0)}\frac{(x-x_1)}{(x_j-x_1)}\cdots\frac{(x-x_{j-1})}{(x_j-x_{j-1})}\frac{(x-x_{j+1})}{(x_j-x_{j+1})}\cdots\frac{(x-x_k)}{(x_j-x_k)} \\ &= \prod_{\substack{0\leq m\leq k \\ m\neq j}} \frac{x-x_m}{x_j-x_m} \end{aligned}$$

Notice that the numerator $\prod_{m\neq j}(x-x_m)$ has k roots at the nodes $\{x_m\}_{m\neq k}$ while the denominator $\prod_{m\neq j}(x_j-x_m)$ scales the resulting polynomial so that $l_j(x_j) = 1$.

The Lagrange interpolating polynomial for those nodes through the corresponding values $\{y_0, y_1, \ldots, y_k\}$ is the linear combination

$$L(x) = \sum_{j=0}^{k} y_j l_j(x),$$

Each basis polynomial has degree k, so the sum $L(x)$ has degree $\leq k$, and it interpolates the data because

$$L(x_m) = \sum_{j=0}^{k} y_j l_j(x_m) = \sum_{j=0}^{k} y_j \delta_{mj} = y_m,$$

Theorem 29.10 (Extension of Lagrange's Theorem to Non Symmetric Functions) *If two rational functions of any set of variables are such that one remains unchanged by all the substitutions of the group to which the other belongs, the first is expressible by means of the second in the form of an integral polynomial whose coefficients are rational symmetric functions of the variables.*

Another statement of Lagrange's Theorem:

If a rational function $\phi(x_1, x_2, \ldots, x_n)$ remains unaltered by all the substitutions which leaves another rational function $\psi(x_1, x_2, \ldots, x_n)$ unaltered, then ϕ is a rational function of ψ and $c_1, c_2, \ldots, c_n$. Note that the group, G to which ϕ belongs contains the group, H to which ψ belongs, i.e. $H \subseteq G$. G is the wider group. Also ϕ is therefore the more symmetric function – a wider set of substitutions keep it unaltered.

Proof

$$\begin{array}{cccc|c|c} I & h_2 & \dots & h_P & \psi = \psi_1 & \phi = \phi_1 \\ g_2 & h_2 g_2 & \dots & h_P g_2 & \psi_{g_2} = \psi_2 & \phi_{g_2} = \phi_2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ g_\nu & h_2 g_\nu & \dots & h_P g_\nu & \psi_{g_\nu} = \psi_\nu & \phi_{g_\nu} = \phi_\nu \end{array} \tag{39}$$

The $\psi_1, \psi_2, \dots, \psi_\nu$ are all distinct; but $\phi_1, \phi_2, ...\phi_n u$ need not be distinct since ϕ belongs to a group G which may be bigger than H. Under any substitution s on $x_1, x_2, \dots, x_n$, the functions $\psi_1, \psi_2, \dots, \psi_\nu$ are merely permuted. Moreover, if s replaces ψ_i by ψ_j, it replaces ϕ_i by ϕ_j. Set (using the Lagrange Interpolation formula)

$$\begin{aligned} g(t) &= (t-\psi_1)(t-\psi_2)\dots,(t-\psi_\nu), \\ \lambda(t) &= g(t)\left(\frac{\phi_1}{t-\psi_1} + \frac{\phi_2}{t-\psi_2} + \dots + \frac{\phi_\nu}{t-\psi_\nu}\right) \end{aligned}$$

so that $\lambda(t)$ is an integral function of degree $\nu - 1$ in t. Since $\lambda(t)$ remains unaltered under every substitution of s, its coefficients are rational symmetric functions of $x_1, x_2, \dots, x_n$ and hence are rational functions of the elementary symmetric functions.

Now we're assuming here that $x_1, x_2, \dots, x_n$ denote indeterminant quantities since $\psi_1, \psi_2, \dots, \psi_\nu$ are algebraically distinct, so that $g'(\psi)$ is not idendically zero. In the case where special values are assigned to $x_1, x_2, \dots, x_n$ such that two or more of the functions $\psi_1, \psi_2, \dots, \psi_\nu$ become numerically equal, then $g'(\psi) = 0$, and ϕ is not a rational function of $\psi, c_1, ..., c_n$. Assuming that we're working with indetermiant $x_1, x_2, \dots, x_n$, and taking $\psi_1 = \psi$ for t, we get

$$\lambda(\psi_1) = [(\psi_1 - \psi_2)(\psi_1 - \psi_3)\dots(\psi_1 - \psi_\nu)]\,\phi_1$$

and in general

$$\phi = \frac{\lambda(\psi)}{g'(\psi)}.$$

Corollary 29.11 *A function can always be found in terms of which any number of given functions can be rationally expressed.*

The groups if the given functions have always one sub-group common to all, for the identity $S = 1$ is common to all. Therefore the functions can all be expressed in terms of any one of the functions peculiar to the common sub-group. Therefore if $\phi, \psi, \xi, \dots,$ are the given functions, then

$$f = \alpha\phi + \beta\psi + \gamma\xi + \dots,$$

where $\alpha, \beta, \gamma, \dots,$ are arbitrary constants, is one such function for the common subgroup ; for any substitution which lease it unaltered must leave $\phi, \psi, \xi, \dots$, unaltered and must be common to the groups of $\phi_1, \psi_1, \xi_1, \dots$.

Corollary 29.12 *Any rational function whatever can be rationally expressed in terms of a function having $n!$ distinct values ; in particular the Galois function.*

The Group of an $n!$-valued function, is included as a sub-group of every other permutation group.

Corollary 29.13 *The roots themselves can be expressed rationally in terms of the Galois function.*

Proof Let $f(x) = 0$ be the equation whose roots are $x_1, x_2, \dots, x_n$, all supposed unequal; and let ψ_1 be a known value of a a rational function ψ of the roots which has $n!$ distinct values.

If all the roots except x_1 be permuted in every possible way, we obtain $\mu = (n-1)!$ distinct values of ψ given by the equation

$$F(x) = (\psi - \psi_1)(\psi - \psi_2)\dots(\psi - \psi_\mu) = 0.$$

The coefficients of this equation when expanded are symmetric functions of $x_2, x_3, \dots, x_n$ and therefore can be expressed in terms of the coefficients of

$$\frac{f(x)}{x - x_1} = 0,$$

and will involve x_1 in a rational form along with the coefficients of $f(x)$ If the expanded equation be represented by $F(\psi, x_1) = 0$, we have $F(\psi_1, x_1) = 0$, since it is satisfied by $\psi = \psi_1$; also we have $f(x_1) = 0$, from which it follows that the equation $f(x) = 0$ and $F(\psi_1, x) = 0$ have a common root. It is easily seen that this is the only common root. If therefore we seek the greatest common divisor of $f(x)$ and $F(\psi_1, x)$, as all the remainders vanish for $x = x_1$ in particular the remainder of the first degree in x of $f(x)$, in which the expression ψ_1 may be replaced by $\psi_2, \psi_3, \dots \psi_\mu$ without altering its value.

Corollary 29.14 *All the values of the Galois function can be expressed rationally in terms of any one of them.*

Proof For all of these values belong to the same group, i.e., the identity.

30 Cauchy's 1815 Theorem

Cauchy's 1815 theorem says that if a function of n variables takes more than two distinct values when you permute the variables, then it must take at least p distinct values, where p is the largest prime less than or equal to n. The number of distinct values always divides n factorial. Symmetric functions take one value, and the only way to get exactly two are by use of the alternating functions, like the Vandermonde product.

Theorem 30.1 (Cauchy's theorem:) *For a function of n variables, $f = f(x_1, \dots, x_n)$, the number of distinct values under all permutations of the variables is either 1, 2, or at least p, where p is the largest prime less than or equal to n.*

Proof Let f be a function of n (distinct) variables $f = f(x_1, \dots, x_n)$. Permuting the variables produces up to $n!$ possible inputs, but we assume f to take only R distinct values under these permutations. By Lagrange's theorem $n!$ is divisible R. Assume for contradiction purposes that $2 < R < p$. Consider a p-cycle σ (a cyclic permutation of order p on p of the variables, fixing the rest). The sequence of function values obtained by applying successive powers of σ to a fixed initial ntuple of variables:

$$f(x),\ f(\sigma(x)),\ f(\sigma^2(x)), \dots,\ f(\sigma^{p-1}(x)).$$

There are at most p distinct values in the above list (since σ^p is the identity). Let

$$v_k = f(\sigma^k(x)) \text{ for } k = 0, 1, ..., p-1,$$

stand for these p values and lest us apply the **Pigeonhole Principle** (i.e., if you have more pigeons than pigeonholes, at least one hole has more than one pigeon). These p values are drawn from the set of R assumed distinct possible values that f can ever take, and we assume $R < p$. These values satisfy the cyclic relation: $v_{k+1} = f(\sigma^{k+1}(x))$ is obtained from v_k by applying $\sigma \pmod p$. By the pigeonhole principle at least two of these p values must be equal.

$$v_r = v_s \text{ for some } 0 \leq s < r \leq p - 1.$$

Let $d = r - s$ (so $1 \leq d \leq p - 1$).

Now apply powers of σ repeatedly. Because σ shifts the indices cyclically (i.e., applying σ moves v_k to $v_{(k+1) \pmod p}$), the assumption $v_r = v_s$ means the sequence is periodic with a smaller step. The equality propagates under shifts: Apply σ any number of times to both sides of $v_r = v_s$. This gives: $v_{r+m} = v_{s+m \pmod p}$ for any integer m. In other words, shifting the indices by the same amount preserves the equality. The index differences are multiples of d: The indices where we know equalities hold are exactly the index positions that differ by multiples of $d \pmod p$. That is, for any integer m: $v_k = v_{(k+m-d) \pmod p}$, whenever the equality chain reaches there.

Primality of p makes d a generator: Crucially because p is prime and d is between 1 and $p-1$, we have $\gcd(d, p) = 1$. Therefore, d has a multiplicative inverse modulo p, and the multiples

$$\{0 \cdot d, 1 \cdot d, 2 \cdot d, \ldots, (p-1) \cdot d\} \pmod p$$

hit every residue class modulo p exactly once and we cycle through all residues modulo p. Starting from any fixed index k, we can reach any other index l by adding a suitable multiple of d modulo p. Therefore $v_k = v_l$ for all k, l. In group theory terms: the subgroup generated by d modulo p is the entire cyclic group $\mathbb{Z}/p\mathbb{Z}$. Therefore entire orbit of $f(\sigma^k x)$ for $k = 0, 1, 2, \ldots, p-1$ collapses to a single value.

Conclusion:

The shifts by multiples of d generate the entire cyclic group of order p. It follows that all the v_k are equal (the orbit collapses to a single value). In particular, $v_0 = v_1$, i.e., $f(x) = f(\sigma(x))$. Since the initial ntuple x was arbitrary, this holds for every starting assignment: σ leaves f invariant (does not change its value). By choosing different p-cycles (or by conjugacy/relabeling of variables), this applies to all p-cycles in S_n.

This implies that all p-cycles leave f invariant (fix every value). One can then construct a 3-cycle as a product (or commutator) of two such p-cycles. Thus, 3-cycles also fix values. A 3-cycle is a product of two transpositions. Analysis of transpositions then shows that either f is fully symmetric ($R = 1$) or it takes exactly two values (like the sign/alternating case, e.g., Vandermonde product). But we assumed $R > 2$, a contradiction. You can't get three or more without hitting at least p values. ∎

Thus, no such R with $2 < R < p$ is possible. The pigeonhole step forces p-cycles (and then smaller cycles) to preserve values, collapsing the possibilities to $R = 1$ or 2. This is the heart of Cauchy's 1815 argument (sometimes phrased in terms of"forms" or "arrangement" of the variables). Modern presentations often frame it in terms of the action of the symmetric group S_n on the values, with stabilizers and orbits (see the **Orbit-Stabilizer Theorem**) but the core pigeonhole idea on powers of a cycle remains the same.

Why Primality of p Is Essential

If p were composite, say d shares a common factor with p, the multiples of $d \pmod p$ would only generate a proper subgroup, and you might only force equality within cosets rather than full invariance under σ. The primality ensures the generated subgroup is the whole group, forcing full invariance. This is the precise bridge from the pigeonhole collision to invariance under the full p-cycle. Once you have invariance under all p-cycles, the rest of the proof proceeds by constructing smaller cycles (e.g., 3-cycles) as products/commutators and analyzing transpositions.

31 Wantzel's proofs

Theorem 31.1 *Every two valued function of n variables is of the form $S_1 \pm S_2\sqrt{\triangle}$, where S_1 and S_2 are integral symmetric functions, and $\triangle$ is the discriminant.*

Proof A two-valued function must belong the the group of order $\frac{1}{2}N$. The only group of this order is the alternating group to which the function $\sqrt{\triangle}$ belongs. The theorem therefore follows as an immediate consequence of the fundamental theorem Thm (29.9).

An Second Proof

Let two values of the function be denoted by ϕ_1 and ϕ_2 and let G_1 and G_2 be the corresponding groups, each of order $\frac{1}{2}N$. In the first place, these two groups must be identical; for it any substitution S of G_1 were to change ϕ_2 to its second value ϕ_1, then S^{-1} would change ϕ_1 into ϕ_2; but this is impossible, since S^{-1} and S each belong to G_1. Every substitution therefore of G_1 must belong to G_2 and vice-versa.

To show now that these groups coincided with the alternating group, consider the function $\psi = \phi_1 - \phi_2$. Any substitution which belongs to the common group leaves this unaltered; any other will change phi_1 to ϕ_2 and ϕ_2 to ϕ_1, and will therefore change the sign of ψ; some transposition, $(x_\alpha x_\beta)$ for example, will have this affect, for no group can include all the transpositions without coinciding with the symmetric group. It is easily inferred that $\psi_1 - \psi_2$ is divisible by $x_\alpha - x_\beta$, and hence by the product of all the differences, since ψ^2 is symmetric.

The quotient of ψ by $\sqrt{\triangle}$ is symmetric. To prove this, let $(\sqrt{\triangle})^m$ be the highest power of $\sqrt{\triangle}$ which occurs in ψ. The quotient of ψ by $(\sqrt{\triangle}^m$ is symmetric, since, if not, it would be an alternating function, and would again contain $\sqrt{\triangle}$ as a factor, which is contrary to assumption. It follows immediately that m is an odd number and that the quotient of ψ by $\sqrt{\triangle}$ is symmetric. Writing therefore $\psi_1 - \psi_2 = A\sqrt{\triangle}$ and $\phi_1 + \phi_2 = B$, where A and B are both symmetric, we have

$$\phi_1 = S_1 + S_2\sqrt{\triangle} \text{ and } \phi_2 = S_1 - S_2\sqrt{\triangle},$$

where S_1 and S_2 are both symmetric functions of the variables $x_1, x_2, \ldots, x_n$. The groups G_1 and G_2 obviously coincide with the group of $\sqrt{\triangle}$ – the Alternating Group.

Theorem 31.2 *Permutations are elements of the Symmetric Group and act on products of functions homomorphically because the action naturally distributes over multiplication. By definition, a permutation rearranges inputs, which respects function evaluation and standard pointwise multiplication. The following operations define the action of the Symmetric Group:*

$$\begin{aligned} \sigma \cdot (f+g) &= \sigma \cdot f + \sigma \cdot g; \\ \sigma \cdot (fg) &= (\sigma \cdot f)(\sigma \cdot g), \\ \tau \cdot (\sigma \cdot f) &= (\tau\sigma) \cdot f \end{aligned}$$

for $\sigma, \tau \in S_n$ and $f, g \in F[x_1, \ldots, x_n)]$. S_n is the set of permutations on n elements $[x_1, x_2, \ldots, x_n]$.

Proof 1. Define the Permutation Action A permutation σ acts on a function f by precomposition [12]:

$$(\sigma \cdot f)(x) = f(\sigma^{-1}(x))$$

2. Evaluate the Action on a Product of Functions We apply the permutation to the pointwise product of two functions, $(fg)(x) = f(x)g(x)$.

$$[\sigma \cdot (fg)](x) = (fg)(\sigma^{-1}(x)).$$

3. Unpack the Function Evaluation Evaluate the product at the permuted input $\sigma^{-1}(x)$:

$$[\sigma \cdot (fg)](x) = f(\sigma^{-1}(x)) \cdot g(\sigma^{-1}(x)).$$

4. Regroup using the Action Definition By the definition of the action in Step 1, we rewrite the scalar outputs:

$$[\sigma \cdot (fg)](x) = (\sigma \cdot f)(x) \cdot (\sigma \cdot g)(x).$$

5. Conclude for the Pointwise Product Since this holds for every element $x \in X$, the functions are equivalent:

$$\sigma \cdot (fg) = (\sigma \cdot f)(\sigma \cdot g).$$

Because this action distributes over the product of functions, it satisfies the required structural property (a homomorphism) for group actions on algebraic structures.

Theorem 31.3 *The alternating functions are the only unsymmetrical functions of n variables of which a power can be symmetric.*

Proof It is sufficient to prove this theorem for prime powers; for if there exists a function $F(x_1, x_2, \ldots, x_n)$ such that F^{pq} is symmetric and p is a prime, then there is also a function $\phi = F^q$ such that ϕ^p is symmetric.

Therefore let

$$\phi^p = S, \text{ a symmetric function.}$$

Since the group of ϕ which is unsymmetric, cannot contain all the transpositions, let $\sigma = (x_\alpha x_\beta)$ be a transposition which converts ϕ into ϕ_j; we have

$$\phi_j^p = \phi^p = S.$$

and therefore $\phi_j = \omega\phi$, where ω is a pth root of unity. Hence

$$\sigma\phi = \phi_j = \omega\phi,$$

and operating again with σ

$$\sigma^2\phi = \omega\sigma\phi = \omega^2\phi,$$

but $\sigma^2 = I$ and therefore $\omega^2 = 1$, and consequently $p = 2$.

Since ϕ^s is symmetric, ϕ is an alternating function. ∎

Theorem 31.4 *For any number, n, of independent elements there is no multiple-valied function of which a power is two-valiued when $n > 4$; and when $n = 3$, or $n = 4$, if there is any such power it is third power.*

Concentrating on prime numbers, and supposing that ϕ is a multiple-valued function whose p power is two-valued we have

$$\phi^p = S_1 + S_2\sqrt{\triangle}. \tag{40}$$

The group of ϕ cannot contain all the circular substitutions of the third order, for if it did, then this group would coincide with the alternating group, and ϕ would be two-valued. Let $\sigma = (x_\alpha, x_\beta, x_\gamma)$ be such a substitutions not contained in the group of ϕ and suppose $\sigma\phi = \phi_j$. From Eq. (40), since $S_1 + S_2\sqrt{\triangle}$ is unaltered by σ, we have

$$\phi^p = \phi_j^p,$$

hence $\phi_j = \omega\phi$, where ω is a p^{th} root of unity. Operating again twice in succession with σ we obtain

$$\begin{aligned} \sigma\phi &= \omega\phi, \\ \sigma^2\phi &= \omega\sigma\phi = \omega^2\phi, \\ \sigma^3\phi &= \omega^2\sigma\phi = \omega^3\phi, \end{aligned}$$

whence, since $\sigma^3 = 1$ (three cycle), we have $\omega^3 = 1$, and therefore $p = 3$.

Again, when the number of elements is greater than 4, there are circular substitutions of the fifth order, and those cannot all be contained in the group of ϕ (or else it will be two-valued and belong to the Alternating Group). Let τ be one of those not contained in this group and $\tau\phi = \phi_j$. We have, as before, from Eq. (40), by applying this substitution (which does not affect the right-hand side),

$$\phi^p = \phi_j^p = S_1 + S_2\sqrt{\triangle}.$$

Hence proceeding as before, we have $\tau\phi = \omega\phi$ and oeprating again on this and the succeeding equations with τ we easily find $\tau^5\phi = \omega^5\phi$; whence $\omega^5 = 1$, since $\tau^5 = 1$, and it is proved that $p = 5$. Now this results being incompatible with the previous value of p obtained, we infer that when the number of elements is greater than 4, it is impossile to find any multiple-valiued function ϕ, a prime power of which will be two-valued.

There are actually, when n is not greater than 4, multiple-valiued functions, a third power of which is two-valued.

32 Abel

After clearing the denominators to ensure all algebraic expressions are expressed as polynomials in the radicals, Niels Henrik Abel's 1826 proof proceeds through a structured combination of field extensions, permutation group properties, and a final proof by contradiction. Abel did not use modern Galois group vocabulary, but his strategy directly mapped out what we now call the Galois theory of fields.

1. The Structure of an Algebraic Solution Abel begins by formalizing what a general "solution by radicals" must look like. If a general quintic equation is solvable, its roots must be expressible by a finite chain of algebraic operations. He structures the general form of a root as a polynomial in a radical:$x = q_0 + q_1R^{1/m} + q_2R^{2/m} + \cdots + q_{m-1}R^{(m-1)/m}$ Here, m is a prime number, and the quantities q_i and R are either rational functions of the original coefficients, or they belong to a lower "order" field extension previously built up by adjoining other radicals.

2. Formulating Abel's Theorem on Radical Functions The core pivot of the proof relies on a fundamental theorem Abel establishes about the algebraic dependencies of these radicals. He proves that if a root of a general quintic can be written in the form above, then the radical itself, $R^{1/m}$, must be expressible as a rational function of the roots of the quintic and the coefficients. By applying permutations to the roots of the general quintic, Abel studies how the value of $R^{1/m}$ behaves. Because the coefficients of the general quintic are independent, independent variables, permuting the roots causes $R^{1/m}$ to cycle through its m distinct algebraic values (the m-th roots of unity multiplied by the radical).

3. Restricting the Prime Index m Abel analyzes what values the prime degree m of the outermost radical can take.A function of 5 independent variables can only take a specific number of distinct values when its variables are permuted. By invoking Cauchy's earlier work on the number of values a symmetric or partially symmetric function can take, Abel shows that the number of values a rational function of 5 variables can take under permutation must be a divisor of $5! = 120$. He narrows down the possible values of the prime index m. The only primes that can mathematically divide these algebraic combinations for 5 variables are $m = 2$ and $m = 5$.

4. Bounding the Value Multiplicity (The Cauchy Constraint) Abel then looks at a critical theorem proved by Augustin-Louis Cauchy: a non-symmetric function of 5 variables cannot take fewer than 5 distinct values unless it takes exactly 2 values (which corresponds to the alternating group A_5, generated by the square root of the discriminant). This creates a rigid algebraic trap:If $m = 5$, the radical expression must take exactly 5 distinct values under root permutations.Abel shows that the only way a function of 5 variables can take exactly 5 values under permutation is if it behaves like a specific linear combination of the roots (similar to a Lagrange resolvent).

5. The Final Contradiction Abel explicitly sets up the equations for these 5-valued functions. He assumes a root looks like:$x = q_0 + q_1 R^{1/5} + q_2 R^{2/5} + q_3 R^{3/5} + q_4 R^{4/5}$ By examining the algebraic behavior of the individual terms under 3-cycle and 5-cycle permutations, he derives a system of equations for the components q_i. He proves that for the general quintic, it is impossible for these equations to hold simultaneously. The permutations of the 5 independent roots require the individual components to vanish or collide in a way that contradicts the assumption that x is a root of a general equation with independent coefficients. Because the assumption that a general root can be expressed as a chain of radicals leads directly to an algebraic impossibility, Abel concludes that the general quintic equation cannot be solved by radicals.

Summary of Abel's Path After Clearing Denominators

Defines Root Anatomy: Expresses the hypothetical root as a polynomial of a prime-degree radical $R^{1/m}$.

Inverts the Radical: Proves that $R^{1/m}$ must be a rational function of the quintic's roots.

Applies Permutations: Analyzes how $R^{1/m}$ changes values when the roots are permuted.

Invokes Cauchy's Bound: Restricts the possible values of m to 2 or 5 based on the combinatorics of 5 variables.

Forces Contradiction: Demonstrates that the algebraic conditions required to form a 5-valued function cannot be met by the coefficients of a general quintic.

To understand how Augustin-Louis Cauchy's work on permutations constrained Niels Henrik Abel, we have to look at the combinatorics of switching variables.
When Abel proved that the outermost radical of a hypothetical solution, $v = R^{1/m}$, must be a rational function of

the five roots $(x_1, x_2, x_3, x_4, x_5)$, he created a bridge between algebra and combinatorics. Because the coefficients of a general quintic are completely independent, permuting the roots creates a set of distinct values for that function. Cauchy's theorem placed a strict mathematical ceiling on exactly how many values such a function could take, narrowing Abel's options down to a dead end.

1. The Core of Cauchy's Theorem (1815) In 1815, Cauchy published a groundbreaking theorem concerning symmetric and non-symmetric functions. He proved a rigid restriction for functions of n independent variables

Theorem 32.1 (Cauchy's Number of Values Theorem:) *If a non-symmetric function of n independent variables changes its value when the variables are permuted, the number of distinct values it can take (its index) must be at least the largest prime factor of n. Furthermore, for $n = 5$, a function cannot have 3 or 4 distinct values. It can only have 1 (fully symmetric), 2, 5, or more (up to $5! = 120$) distinct values.*

2. How This Restricted Abel's Prime Radical (m) Abel assumed the general root of the quintic could be written as a polynomial in a radical of prime degree m:$x = q_0 + q_1 v + q_2 v^2 + \cdots + q_{m-1} v^{m-1}$ where $v = R^{1/m}$ By definition, if you multiply v by the m-th root of unity (ω), its m-th power R stays exactly the same. Therefore, the function $v(x_1, x_2, x_3, x_4, x_5)$ can take exactly m distinct values as its variables are permuted. Abel applied Cauchy's theorem to this m:Because v takes exactly m values, Cauchy's theorem dictates that m must be a possible number of values for a 5-variable function. According to Cauchy, for 5 variables, the number of values can only be 1, 2, 5, or ≥ 6. Since m must be a prime number by algebraic design, the only prime numbers allowed in that set are $m = 2$ and $m = 5$. Cauchy's theorem instantly destroyed the possibility of using cubic roots ($m = 3$), seventh roots ($m = 7$), or any higher prime radicals at the final stage of a quintic solution.

3. Eliminating the $m = 2$ Option An $m = 2$ radical means a square root. If $m = 2$, the function v takes exactly 2 values under permutation. Cauchy's work had already thoroughly mapped out 2-valued functions of 5 variables. The only way a function of 5 variables can take exactly 2 values is if it is a symmetric function multiplied by the alternating sign function (the square root of the discriminant, $\sqrt{\Delta}$). Abel knew that adjoining a square root ($\sqrt{\Delta}$) merely splits the symmetric group S_5 into the alternating group A_5. It simplifies the problem slightly but does not solve the equation. A square root alone cannot express five distinct, independent roots. Therefore, $m = 2$ could not be the final step.

4. Trapping the $m = 5$ Option This left Abel with exactly one remaining choice: $m = 5$. The outermost radical had to be a fifth root, meaning the function v must take exactly 5 distinct values when the roots are permuted. Abel turned back to Cauchy's structural analysis of 5-valued functions. Cauchy had proven that if a function of 5 variables has exactly 5 values, the permutations that leave the function unchanged must form a specific subgroup of order 24 (isomorphic to the symmetric group S_4). To satisfy this rigid combinatorial structure, Abel showed that the function v must be a linear combination of the roots structured precisely like a Lagrange resolvent:$v = x_1 + \alpha x_2 + \alpha^2 x_3 + \alpha^3 x_4 + \alpha^4 x_5$(where α is a fifth root of unity).

5. The Closing of the Trap By using Cauchy's theorem to force v into this exact, unyielding algebraic shape, Abel was able to execute his final blow. He applied a simple 3-cycle permutation (e.g., swapping $x_1 \to x_2 \to x_3 \to x_1$) to the roots. Because a 3-cycle permutation belongs to the alternating group A_5 but not to the subgroup that keeps a 5-valued function stable, it forced a contradiction. Abel proved that under these required permutations, the algebraic coefficients would have to simultaneously equal completely different values, meaning the assumed fifth-root function could not actually exist for independent variables. Without Cauchy's theorem bounding the permissible number of values to 2 and 5, Abel would have had to fight an infinite number of potential radical degrees ($m = 3, 5, 7, 11 \ldots$). Cauchy provided the boundaries that allowed Abel to corner the quintic and prove it impossible.

In his 1826 paper, Niels Henrik Abel used a beautifully simple property of 3-cycle permutations to snap his algebraic trap shut. He didn't use modern group theory notation (like A_5 or S_5), but he expressed the exact same logical mechanism. Here is the exact step-by-step breakdown of how a single 3-cycle permutation created his final, fatal contradiction.

1. The Setup: The Five-Valued RadicalThrough Cauchy's theorem, Abel established that the final, outermost radical needed to solve the quintic must be a fifth root, which we can call v:$v = R^{1/5}$ Because v is a fifth root, permuting the roots of the quintic can only change v by multiplying it by some fifth root of unity, ω (where $\omega^5 = 1$). Therefore, under the full group of permutations, v can take exactly 5 distinct values:$\{v, \omega v, \omega^2 v, \omega^3 v, \omega^4 v\}$

2. Applying the 3-Cycle Permutation Abel then chose an arbitrary 3-cycle permutation of the roots, which we can denote as p. A 3-cycle simply takes three roots and rotates them in a loop (for example, swapping $x_1 \to x_2 \to x_3 \to x_1$), leaving the other two roots completely untouched. Let's apply this 3-cycle p to our radical function v. As established, applying any permutation to the roots can only multiply v by some fifth root of unity, say ω^k:$p(v) = \omega^k \cdot v$

3. The Power of Order Three A 3-cycle has a strict geometric property: if you execute it exactly three times, every element returns to its original starting position ($p^3 = e$, the identity permutation). Abel applied the exact same 3-cycle permutation three times in a row to both sides of the equation:

First time: $p(v) = \omega^k \cdot v$

Second time: $p(p(v)) = \omega^k \cdot \omega^k \cdot v = \omega^{2k} \cdot v$

Third time: $p(p(p(v))) = \omega^k \cdot \omega^k \cdot \omega^k \cdot v = \omega^{3k} \cdot v$

Because applying a 3-cycle three times restores the roots to their original order, the left side must evaluate back to the original, un-permuted value of v:$v = \omega^{3k} \cdot v$ For this equation to hold true, the multiplier must equal 1:$\omega^{3k} = 1$

4. The Arithmetic Lock We now have two strict rules dictating the behavior of this multiplier:From the nature of the fifth root, ω is a fifth root of unity ($\omega^5 = 1$). From the nature of the 3-cycle, ω^{3k} must equal 1. Because 3 and 5 are prime numbers that share no common factors (coprime), the only power that satisfies both conditions simultaneously is $k = 0$, meaning $\omega^0 = 1$. Therefore:$p(v) = 1 \cdot v = v$ This reveals an astonishing fact: every single 3-cycle permutation leaves the radical v completely unchanged.

5. The Final Contradiction This is where Abel applies the coup de gräce using Cauchy's framework:

Fact A: In combinatorics, the 3-cycle permutations are the foundational building blocks that generate the entire **Alternating Group (A_5)**. Because v is unaffected by every 3-cycle, it is completely invariant under the entire alternating group.

Fact B: According to Cauchy's structural theorems, any function of 5 variables that is completely invariant under the alternating group can take **at most 2 distinct values** across all possible permutations (acting like the square root of a discriminant).

Now, look at the two numbers we have forced upon the exact same function v: From its definition as a fifth root, v must take exactly 5 values. From the 3-cycle arithmetic, v can take at most 2 values. A function cannot simultaneously possess exactly 5 values and at most 2 values. The assumption that a general quintic can be solved by a chain of radicals containing a fifth root collapses into an absolute algebraic impossibility.

33 The Commutator Subgroup (The Abelian Quotient)

Definition Let G be a group. An element $g \in G$ is called a commutator if $g = aba^{-1}b^{-1}$ for elements $a, b \in G$. The smallest subgroup that contains all commutators of G is called the commutator subgroup or derived subgroup of G, and is denoted by G'.

A highly practical relationship emerges with the commutator subgroup G'. The quotient G/G' forms the largest Abelian (commutative) group that can be created from G. If you "mod out" by the commutator subgroup, all order-dependent operations are eliminated.

Proposition 33.1 *Let G be a group with commutator subgroup G'.*

(a) The subgroup G' is normal in G, and the factor group G/G' is Abelian.

(b) If N is any normal subgroup of G, then the factor group G/N is Abelian, if and only if $G' \in N$.

Proof (a) Let $x \in G'$ and let $g \in G$. Then $gxg^{-1}x^{-1} \in G'$, and since $x \in G'$, we must have $gxg^{-1} = gxg^{-1}x^{-1}x \in G'$.

The factor group G/G' must be Abelian since for any cosets aG', bG', because G' is normal, we have $xG' = G'x$ for all $x \in G$.

$$aG'bG'a^{-1}G'b^{-1}G' = abG'G'a^{-1}b^{-1}G'G' = aba^{-1}b^{-1}G' = G'.$$

Proof (b) Let N be a normal subgroup of G. If $N \supseteq G'$, then G/N is a homomorphic image of G/G' and must be Abelian. Conversely, suppose that G/N is Abelian. Then $aNbN = bNaN$ for all $a, b \in G$, or simply $aba^{-1}b^{-1}N = N$, showing that every commutator of G belongs to N. This implies $G' \subseteq N$. ∎

Theorem 33.2 *The Symmetric Group G on n letters is not solvable unless $n \leq 4$.*

Proof Let $G = S_0 \supseteq S_1 \supseteq S_2 \supseteq \ldots \supseteq S_s$ be any chain of subgroups, each normal in the preceeding with cyclic quotient-group S_{k-1}/S_k; we shall prove by induction on s that S_s must contain every 3-cycle (ijk). This will imply that $S_s > I$, and so, that G cannot be solvable. Since $S_0 = G$ contains every 3-cycle, it is sufficient to show that if S_{k-1} contains every 3-cycle, then so does S_k. First note that if the permutations ϕ and ψ are both in S_{k-1}, then their so-called commutator $\gamma = \phi^{-1}\psi^{-1}\phi\psi$ is in S_s. To see this, consider their images ϕ' and ψ' in S_{k-1}/S_k. The latter, being cyclic, is commutative; hence

$$\gamma' = \phi'^{-1}\psi'^{-1}\phi'\psi' = \phi'^{-1}\psi'^{-1}\psi'\phi' = I \text{ in } S_{k-1}/S_k,$$

which implies that $\gamma \in S_k$. But in the special case when $\phi = (ijk)$ and $\psi = (jkm)$, where i, j, k are given and l, m are any two other letters (such letters exist unless $n \leq 4$), we have

$$\gamma = (jli)(mkj)(ilk)(jkm) = (ijk) \in S_k, \text{ for all } i, j, k.$$

This proves that S_k contains every 3-cycle, as desired.

The quotient chain doesn't terminate on the Identity, therefore the group G is not solvable unless $n \leq 4$. The 3-cycles generate the Alternating group which is simple and has no normal subgroups.

34 Hamilton

In 1836, the brilliant Irish mathematician Sir William Rowan Hamilton read Niels Henrik Abel's 1826 paper and found parts of the proof to be brilliant but "obscure" and "problematic". Recognizing gaps in how Abel handled permutations and nested radical operations, Hamilton published a massive, highly meticulous 130-page refinement in 1839 entitled "On the Argument of Abel". Hamilton's core contribution was cleaning up the 3-cycle permutation logic. He realized that Abel had jumped to conclusions regarding how nested radicals behave when root transformations occur. Hamilton rebuilt the 3-cycle argument into a completely bulletproof algebraic mechanism.

1. Identifying Abel's Flaw: The Hidden "Ambiguity" Abel had asserted that because a radical $v = R^{1/5}$ must take 5 distinct values under permutation, applying a 3-cycle permutation must multiply it by some fifth root of unity ω^k. Hamilton pointed out a deep issue here: radicals are multi-valued expressions. When you apply a permutation to the roots of a polynomial, you alter the underlying variables. You cannot simply assume that the radical function tracks to the exact same branch of the root without a rigorous tracking mechanism. Hamilton realized that if the radical inside the function was nested under other radicals, a permutation might swap the branches of the inner radicals, destroying Abel's straightforward equation $p(v) = \omega^k v$.

2. Hamilton's Fix: Commutators of 3-Cycles To resolve this branch-switching ambiguity, Hamilton bypassed simple 3-cycles and introduced what modern group theorists call commutators. Instead of just checking what a single 3-cycle does to a radical, Hamilton looked at the interaction between two different 3-cycles, which we can call P and Q. He analyzed the specific composite operation:Commutator $= P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ Hamilton proved a stunning combinatorial property of five variables: you can construct any 3-cycle permutation of 5 elements by combining the commutators of other 3-cycles. For example, let:P be the 3-cycle swapping roots $(1, 2, 3)$$Q$ be the 3-cycle swapping roots $(3, 4, 5)$ If you apply them in the sequence $P \to Q \to P^{-1} \to Q^{-1}$, the net result is a completely new 3-cycle permutation.

3. Neutralizing the Branch-Switching Problem Why did Hamilton use this complex, four-step sequence instead of Abel's single 3-cycle? Because the commutator structure cancels out any branch-switching ambiguities of nested radicals. Let's look at how Hamilton applied this to the 5-valued radical v: Suppose permutation P multiplies v by some root of unity and shifts an inner branch. Suppose permutation Q multiplies v by another root of unity and shifts an inner branch differently. When you run the sequence in reverse (P^{-1} then Q^{-1}), the operations on the internal branches perfectly unwind and cancel each other out, because the underlying field extensions are Abelian. However, the fifth root of unity multipliers are just simple complex numbers, which are commutative ($\omega^a \cdot \omega^b = \omega^b \cdot \omega^a$). Therefore, the multipliers track through the commutator like this:

$$v \xrightarrow{P} \omega^a v \xrightarrow{Q} \omega^b \omega^a v \xrightarrow{P^{-1}} \omega^{-a} \omega^b \omega^a v \xrightarrow{Q^{-1}} \omega^{-b} \omega^{-a} \omega^b \omega^a v$$

Because complex multiplication commutes, the powers completely annihilate one another:

$$\omega^{-b} \omega^{-a} \omega^b \omega^a = \omega^{-b-a+b+a} = \omega^0 = 1$$

4. The Refined Contradiction Hamilton's math forced a beautiful trap:

By Combinatorics: The commutator sequence $PQP^{-1}Q^{-1}$ successfully creates a brand new, non-trivial 3-cycle permutation that alters the roots.

By Algebra: The exact same commutator sequence reduces the radical's multiplier to exactly 1, meaning the radical v is left entirely unchanged.

Hamilton proved that every single 3-cycle permutation must leave the value of the radical v completely invariant, regardless of how many nested, lower-order radicals were sitting underneath it. This completely sealed the mathematical loophole in Abel's paper. Since the 3-cycles generate the alternating group A_5, the radical v cannot possess 5 distinct values, and the proof of the quintic's insolvability was officially completed with total rigor.

To map out the algebraic field extensions with total precision and prevent any hidden multi-valued ambiguity, Sir William Rowan Hamilton developed an incredibly strict, highly formalized notation system in his 1839 paper, "On the Argument of Abel". Instead of writing standard radical signs like $\sqrt{x}$ or $\sqrt[5]{R}$ – which he argued were mathematically dangerous because they obscured multi-valued branch-switching – Hamilton used nested levels of accent marks (!, !!, !!!) appended to the variable names, combined with explicit prime indexes.

1. The Base Fields and Independent Variables Hamilton starts with the arbitrary coefficients or roots of the general n-th degree polynomial. He denotes these completely independent variables using basic lowercase subscripts: $a_1, a_2, a_3, \ldots, a_n$ Any function that can be built by simple addition, subtraction, multiplication, or division of these variables is written as a standard rational function:$f_1(a_1, a_2, \ldots, a_n)$

2. First-Order Radicals (The Single Accent !) When the first layer of radicals is adjoined to the base field (e.g., square roots or cube roots of the initial coefficients), Hamilton marks them with a single exclamation point accent (!). He defines a set of n' distinct first-order radicals $a'_1, a'_2, \ldots, a'_{n'}$ by their explicit algebraic relations:

$$\begin{aligned}
(a'_1)^{\alpha'_1} &= f_1(a_1, a_2, \ldots, a_n) \\
(a'_2)^{\alpha'_2} &= f_2(a_1, a_2, \ldots, a_n) \\
&\vdots \\
(a'_{n'})^{\alpha'_{n'}} &= f_{n'}(a_1, a_2, \ldots, a_n)
\end{aligned}$$

Meaning: α'_i is the prime root index (like 2, 3, or 5). The variable a'_1 represents the radical itself, structurally tied to a specific base rational function.

3. Second-Order Radicals (The Double Accent !!) To build the next layer of the tower (radicals containing other radicals inside them), Hamilton adds a second exclamation point (!!). These new radicals are functions of both the base variables and the newly created first-order radicals:

$$(a''_1)^{\alpha''_1} = f'_1(a'_1, a'_2, \ldots, a'_{n'}, a_1, a_2, \ldots, a_n)$$

By forcing the notation to explicitly list every single sub-radical parameter in the parentheses, Hamilton guaranteed that if a permutation altered a lower-tier root (like a'_1), its exact ripple effect onto the higher-tier root (a''_1) could be meticulously traced.

4. How Hamilton Explicitly Wrote a Cubic Root Solution To show how his notation mapped to historical solutions before attacking the quintic, Hamilton explicitly wrote out the classical Cardano solution for the general cubic equation

$$x^3 + a_1x^2 + a_2x + a_3 = 0$$

using his system: He tracked the solution through a strict sequence of rational functions (c_1, c_2) and nested radical levels (a_1', a_1''):

Symmetric Base Functions:

$$\begin{aligned} c_1 &= -\frac{1}{54}(2a_1^3 - 9a_1a_2 + 27a_3) \\ c_2 &= \frac{1}{9}(a_1^2 - 3a_2) \end{aligned}$$

First-Order Radical (Level 1): A square root of the discriminant components:

$$(a_1')^2 = f_1(a_1, a_2, a_3) = c_1^2 - c_2^3$$

Second-Order Radical (Level 2): A cube root that nests the first square root inside it:

$$(a_1'')^3 = f_1'(a_1', a_1, a_2, a_3) = c_1 + a_1'$$

Finally, he expresses the absolute root x as a pure, single-valued algebraic function b:$x = b = f(a_1'', a_1', a_1, a_2, a_3) = -\frac{a_1}{3} + a_1'' + \frac{c_2}{a_1''}$

Why this Notation Settled the Argument Whenever previous mathematicians (including Ruffini) tried to permute the variables in a radical expression, critics argued that equations like $\sqrt{R} \to -\sqrt{R}$ were ambiguous because $\sqrt{R}$ didn't have an inherent, single-valued identity. By replacing standard radical signs with explicitly defined variables (a_1', a_1'') indexed to precise functional domains, Hamilton turned multi-valued radical expressions into a precise, deterministic system of tracking algebraic field elements.

A modern exposition of Hamilton's proof modern exposition translates his cumbersome, 130-page tower of algebraic notation into the streamlined language of modern abstract algebra, field extensions, and topological/geometric tracking. Historically, this presentation bridges the gap between Abel's pure algebraic manipulation and Galois's abstract group theory. A famous contemporary variant of this approach was popularized by the legendary mathematician Vladimir Arnold, who geometricized Hamilton's permutation logic [10].

1. The Setup: Towers of Radical Extensions In modern field theory, solving an equation by radicals means constructing a sequential chain of field extensions (a radical tower). We start with the base field $F_0 = \mathbb{C}(a_1, a_2, \ldots, a_5)$ generated by the independent coefficients of the quintic. A solution implies reaching a final field F_k that contains the roots, where each step in the tower adjoins a prime-degree radical:

$$F_0 \subset F_1 \subset F_2 \subset \cdots \subset F_k$$

Where $F_{i+1} = F_i(\sqrt[m_i]{R_i})$ for some $R_i \in F_i$, and m_i is a prime number.

2. The Tracking Mechanism (Hamilton's Riemann Surfaces) In a modern topological or complex-analytic presentation, we treat the roots $x_1, \ldots, x_5$ as multi-valued continuous functions mapping from the space of coefficients. When we vary the coefficients along a closed loop in complex space, the coefficients return to their exact starting values. However, the roots $x_1, \ldots, x_5$ trace paths that may permute their identities (e.g., executing a 3-cycle like $x_1 \to x_2 \to x_3 \to x_1$). Hamilton's foundational insight was figuring out how the radical tower handles these root loops.

3. The Commutator Loop Let P and Q be two distinct loops in the coefficient space that cause 3-cycle permutations on the roots:Loop P induces the permutation $\sigma = (1\,2\,3)$Loop Q induces the permutation $\tau = (3\,4\,5)$ Now, consider a composite loop known as the commutator loop:$\gamma = P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ If you track the permutation of the roots along the commutator loop γ, the net result is a brand-new 3-cycle permutation: $\gamma \implies \sigma\tau\sigma^{-1}\tau^{-1} = (1\,2\,3)(3\,4\,5)(1\,3\,2)(5\,4\,3) = (1\,3\,4)$

4. The Algebraic Triviality of the Loop Let's look at what happens to the elements inside the radical tower when we travel along this commutator loop γ. The first extension is $F_1 = F_0(\sqrt[m]{R_0})$, where R_0 is a basic rational function of the coefficients. Traveling along any closed loop P or Q can only multiply the radical $v = \sqrt[m]{R_0}$ by some root of unity ω^a or ω^b.

Because complex numbers commute ($\omega^a \cdot \omega^b = \omega^b \cdot \omega^a$), tracking the value of v along the commutator loop $\gamma = PQP^{-1}Q^{-1}$ results in: $\omega^a \cdot \omega^b \cdot \omega^{-a} \cdot \omega^{-b} = \omega^{a+b-a-b} = \omega^0 = 1$. Therefore, traveling along the commutator loop completely un-twists the radical and forces it back to its original branch value. The function v is invariant under γ.

5. Climbing the Tower Hamilton's absolute refinement over Abel was proving that this invariance propagates perfectly up every single level of the tower.

> Suppose every radical up to field F_i returns to its starting value after running a commutator loop of commutators.
>
> Then any new radical adjoined in F_{i+1} will also see its branch-shifts commute and cancel out if the loop is nested deep enough.

In modern group theory language, we are exploiting the fact that the Symmetric Group (S_5) and the Alternating Group (A_5) are perfect groups. Their commutator subgroups do not shrink:

$$[A_5, A_5] = A_5$$

No matter how many times you take commutators of commutators, you can always find a loop sequence that generates a non-trivial 3-cycle permutation on the roots.

6. The Modern Contradiction

> From the Geometry/Topology: We can construct a highly nested commutator loop $\gamma = [[P, Q], [R, S]] \dots$ that still yields a non-trivial 3-cycle permutation on the roots, meaning the roots **must change position**.
>
> From the Algebra: Because the loop is a deeply nested commutator, every single layer of the radical tower is forced to evaluate cleanly back to a multiplier of 1, meaning any formula built of radicals **cannot change value**.

Because a valid formula for a root must change when the root itself changes, the radical formula cannot exist.

Key Takeaway The modern view of Hamilton's proof shows that algebraic solvability requires a group that can be broken down into trivial steps via commutators (a solvable group). Because the root permutations of a quintic

(A_5) constantly regenerate themselves under commutator operations, they create an infinite geometric loop that any finite tower of radicals is structurally incapable of untangling.
To understand why a highly nested commutator loop of 3-cycles never shrinks or vanishes, we have to look at the intersection of topology and group theory. This is the exact mechanism that Vladimir Arnold geometricized in his modern exposition of Abel and Hamilton's work.

The core secret lies in a unique combinatorial property of 5 or more items: the alternating group A_5 is a "perfect group," meaning it is its own commutator subgroup.

1. The Geometric Setup: Loops as Permutations Imagine the five roots of the quintic as five distinct points on a complex plane.

> A loop in the space of coefficients acts like a smooth animation that moves these five points around each other before bringing them back to the exact same five starting slots.
>
> Even though the slots are the same, individual roots have swapped places. Therefore, every continuous loop maps directly to a specific permutation.
>
> A loop that swaps three roots in a circle maps to a 3-cycle.

2. The Mechanics of a Base Commutator $[P, Q]$ Let's see what happens geometrically when we build a simple commutator loop from two different 3-cycle loops, P and Q. Suppose:

> Loop P moves the roots in slots 1, 2, and 3. Its permutation is $\sigma = (1\,2\,3)$.
>
> Loop Q moves the roots in slots 3, 4, and 5. Its permutation is $\tau = (3\,4\,5)$.

Notice that these loops overlap at slot 3. This overlap is where the geometric magic happens. Let's trace a physical point starting in slot 1 through the commutator loop $\gamma = P \cdot Q \cdot P^{-1} \cdot Q^{-1}$:

> Run Loop P: The root in slot 1 moves to slot 2.
>
> Run Loop Q: Loop Q only affects slots 3, 4, and 5. Our root (now in slot 2) sits completely still.
>
> Run Loop P^{-1} (Reverse of P): This moves whatever is in slot 2 back to slot 3. Our root is now in slot 3.
>
> Run Loop Q^{-1} (Reverse of Q): Loop Q^{-1} rotates slots $(5\,4\,3)$ into $(3\,5\,4)$. Therefore, the root sitting in slot 3 gets pushed into slot 4.

By tracking every slot this way, the net result of running this four-step loop sequence is:

$$\sigma\tau\sigma^{-1}\tau^{-1} = (1\,2\,3)(3\,4\,5)(1\,3\,2)(5\,4\,3) = (1\,3\,4)$$

The Geometric Upshot: By combining four loops that only perform 3-cycles, we did not unwind the permutation. Instead, we generated a brand-new, completely non-trivial 3-cycle permutation $(1\,3\,4)$.

3. Climbing the Nesting Ladder: $[[P, Q], [R, S]]$ Because the commutator of two 3-cycles yields another 3-cycle, we can repeat this process infinitely. Let's define two separate commutator loops, each engineered from overlapping 3-cycles:

> Let loop $A = [P, Q]$, which yields the 3-cycle $\alpha = (1\,3\,4)$.

Let loop $B = [R, S]$, which yields the 3-cycle $\beta = (2\,4\,5)$.

Now, we nest them into a deeper commutator loop: $\gamma_{\text{nested}} = [A, B] = A \cdot B \cdot A^{-1} \cdot B^{-1}$ Because A and B are both 3-cycles that overlap at slot 4, running this highly nested loop sequence will, by the exact same algebraic logic, yield another brand-new 3-cycle: $\alpha\beta\alpha^{-1}\beta^{-1} = (1\,3\,4)(2\,4\,5)(1\,4\,3)(5\,4\,2) = (1\,4\,2)$ Even though γ_{nested} is a massive string of 16 individual base loops wrapped inside each other, **the roots are still actively swapping places at the end of the set of permurations**.

4. Why This Fails for Lower-Degree Equations (The Contrast) To see why this topological property traps the quintic, look at what happens if you try this with a cubic equation (3 roots).

The only 3-cycles available for 3 roots are $(1\,2\,3)$ and $(1\,3\,2)$.

If you compute their commutator:$(1\,2\,3)(1\,3\,2)(1\,3\,2)(1\,2\,3) =$ identity (no movement)

For 3 roots (and similarly for 4 roots), if you nest commutators deep enough, the permutations eventually crash into the identity. The loops completely untangle, the roots stop moving, and the contradiction vanishes –which is precisely why cubics and quartics can be solved by radicals.

5. The Trap Closes For 5 or more roots, you can nest commutators 100 levels deep, and you will never run out of overlapping 3-cycles. You can always find a loop sequence where the roots physically swap positions at the end.

Yet, as Hamilton proved, a 100-level-deep nested commutator forces the algebra of any radical formula to cancel its multipliers out to exactly 1, meaning the formula outputs the exact same value it started with.

The topology proves the roots **must move**, while the algebra proves the radical formula **cannot move**. This permanent geometric mismatch is the ultimate proof that a general formula for the quintic cannot exist.

Nesting commutators deepens the loop chain to **force each successive layer of the radical tower to evaluate to a multiplier of 1, effectively neutralizing them one by one**.

In abstract algebra, we call this climbing down the derived series of a group. Geometrically, each level of nesting strips away the multi-valued "twisting" of a specific tier of radicals, exposing the next layer down until the entire formula is forced to act like a single-valued rational function.

1. The Anatomy of a Tiered Radical Tower To see how nesting targets specific levels, we must look at how an algebraic formula is built. Every radical solution is a nested hierarchy of dependencies:

Level 1 Radicals (F_1): Radicals whose inputs are simple, single-valued rational functions of the coefficients (e.g., $v_1 = \sqrt[5]{R_0}$).

Level 2 Radicals (F_2): Radicals whose inputs contain Level 1 radicals inside them (e.g., $v_2 = \sqrt[3]{5 + v_1}$).

Level 3 Radicals (F_3): Radicals whose inputs contain Level 2 radicals (e.g., $v_3 = \sqrt[2]{\cdots + v_2}$).

When you move the coefficients of a polynomial along a single closed loop, **every level of radical can twist simultaneously**. A single loop might multiply v_1 by ω, which shifts the input of v_2, causing v_2 to jump to a completely different Riemann sheet. It looks like chaotic, untrackable behavior.

2. A 1-Level Commutator $[P, Q]$ Targets Level 1 Radicals Suppose we run a simple commutator loop $\gamma_1 = PQP^{-1}Q^{-1}$.

Let's look at a Level 1 radical, $v_1 = \sqrt[m]{R_0}$. Because R_0 is just a basic rational function of the coefficients, it has no multi-valued ambiguity. Loop P can only multiply v_1 by some constant complex root of unity ω^a, and loop Q multiplies it by ω^b.

Because complex numbers commute, tracking v_1 along the 4-step sequence γ_1 gives:$v_1 \to \omega^a v_1 \to \omega^b \omega^a v_1 \to \omega^{-a}\omega^b\omega^a v_1 \to \omega^{-b}\omega^{-a}\omega^b\omega^a v_1 = 1 \cdot v_1$

The Result: The 1-level commutator has completely neutralized Level 1 radicals. After running γ_1, v_1 returns exactly to its starting branch. To Level 1 radicals, the loop γ_1 acts exactly like the identity loop (doing nothing).

3. A 2-Level Nested Commutation $[A, B]$ Targets Level 2 Radicals Now, what happens to a Level 2 radical, $v_2 = \sqrt[k]{f(v_1)}$, when we run that same 1-level commutator γ_1?

Even though v_1 returns to its original value at the end of γ_1, it didn't stay still during the loop. Halfway through the loop, v_1 was multiplied by ω^a. This branch-shift altered the inner function $f(v_1)$, which forced the outer radical v_2 to jump to a different branch. Therefore, at the end of γ_1, v_2 might be multiplied by some root of unity ζ^c.

To fix this, we nest the commutator. We build two separate 1-level commutator loops, A and B.

> As proven above, loop A leaves all Level 1 radicals completely unchanged at its conclusion. It can only multiply v_2 by ζ^c.
>
> Loop B also leaves all Level 1 radicals completely unchanged at its conclusion. It can only multiply v_2 by ζ^d.

Now, we construct the 2-level nested commutator:$\gamma_2 = [A, B] = A \cdot B \cdot A^{-1} \cdot B^{-1}$

Because Level 1 radicals (v_1) evaluate to a multiplier of 1 at the end of both A and B, they act like fixed, single-valued constants relative to these loops. Therefore, the multipliers ζ^c and ζ^d experienced by v_2 are just simple, independent complex numbers. They commute perfectly:$\zeta^c \cdot \zeta^d \cdot \zeta^{-c} \cdot \zeta^{-d} = 1$

The Result: The 2-level nested commutator has now neutralized Level 2 radicals. After running γ_2, both Level 1 and Level 2 radicals are forced back to their starting values.

4. The Inductive Collapse of the Tower This is a perfect inductive trap. Every time you add another layer of commutator nesting, you neutralize the next corresponding tier of the radical tower:

Loop Structure		Neutralized Radicals
Base Loops (P, Q)	$\longrightarrow$	None (All branches twist)
1-Level $[P, Q]$	$\longrightarrow$	Level 1 Radicals return to 1
2-Level $[[P, Q], [R, S]]$	$\longrightarrow$	Level 2 Radicals return to 1
3-Level $[[[P, Q], [R, S]], [[T, U], [V, W]]]$	$\longrightarrow$	Level 3 Radicals return to 1
$\vdots$		$\vdots$

If a quintic formula uses a tower of radicals that goes k layers deep, a k-level nested commutator loop will force every single radical in that entire formula to evaluate back to a multiplier of 1. The entire multi-tiered algebraic expression is stripped of its multi-valued nature and behaves like a rigid, single-valued rational function.

Yet, because the roots of the quintic belong to the perfect group A_5, a k-level nested commutator of 3-cycles still physically permutes the actual roots. The algebra has been frozen solid by the nesting, but the geometry refuses to stop moving—proving that the formula and the roots can never match.

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